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Research paper

A prompting multi-task learning-based veracity dissemination consistency reasoning augmentation for few-shot fake news detection

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ABSTRACT

With the rise of social media, traditional news channels are being supplanted, leading to increased prevalence of fake news. Existing Fake News Detection (FEND) methods, while effective in fully-supervised settings, struggle with data scarcity and lack robustness in few-shot scenarios. Social pattern analysis shows that users tend to share information aligning with their existing beliefs, a behavior known as "Confirmation Bias". This bias influences how news spreads within social groups, where consistent sharing patterns among closely connected users often serve as credible indicators of news authenticity. We refer to this phenomenon as the "user spreading news engagement bias". Inspired by this, we propose a novel **P**rompting **M**ulti-**T**ask **L**earning framework guided by news veracity **Dis**semination **Co**nsistency reasoning augmentation for few-shot FEND, abbreviated as **PMTL-DisCo**, which leverages advanced Artificial Intelligence (AI) techniques, such as multi-task learning (MTL), prompt-based tuning, and masked language model (MLM), for feature extraction and optimization. Unlike prior works that primarily rely on pre-trained language models (PLMs) for feature extraction, PMTL-DisCo integrates the following innovations: (1) an auxiliary task, "news distributed representation optimization," that leverages dissemination consistency signals from neighboring news cases to enhance feature learning. (2) a high-quality, expanded label words-based verbalizer engineering, which improves prompt-tuning performance through adaptive multi-label mappings. (3) a multi-neighbor reasoning enhancement strategy, which refines predictions by incorporating the veracity features of socially connected news cases. These innovations address the limitations of existing FEND methods, particularly their inability to exploit cross-news dissemination patterns effectively. Extensive experiments and ablation studies on three widely used benchmarks demonstrate that PMTL-DisCo outperforms state-of-the-art few-shot FEND models, showcasing its superior generalization and robustness in low-resource settings.

1. Introduction

With the advancement of information technology and the widespread use of the Internet, accessing information and understanding the outside world has become more convenient and efficient for people, significantly enhancing our daily lives (Hu et al., 2022b). Due to the prevalence of fake new on the internet distorting facts, it leads social users to adopt distorted thoughts and incorrect stances, influencing the general populace and even their stance and behavioral decisions (Shu et al., 2017). The rampant spread of Internet fake news is one of the primary insecure factors affecting social stability, bringing

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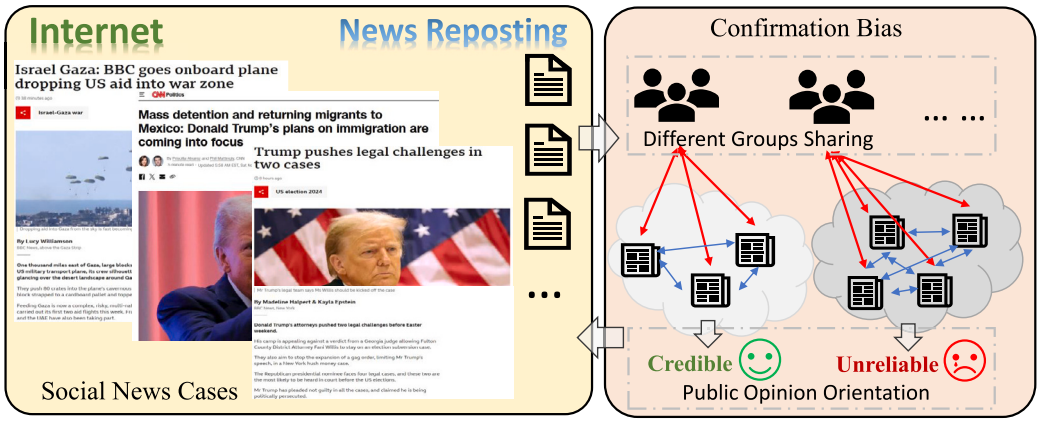


Fig. 1. A visual concept illustration of the social phenomenon: Confirmation Bias.

about significant negative effects that cannot be ignored for the healthy development of society (Elhadad et al., 2019).

Fake News Detection (FEND), also known as disinformation/rumor detection, serves to determine whether a Internet news article is intentionally or unintentionally misleading, regardless of its topic or domain, through automated technology (Shu et al., 2017; Nan et al., 2021). This ensures the timely prevention of the spread of disinformation and rumors within the social ecosystem. Moreover, due to the thriving development of the internet, research into FEND is gaining increasingly attention from both the scientific community and government security agencies.

The mainstream FEND approaches primarily rely on Artificial Intelligence (AI) technologies, particularly machine learning and deep learning (Elhadad et al., 2019; Hu et al., 2022b; Wang et al., 2023, 2021). Recently, with the superior semantic understanding and analysis capabilities of Pre-trained Language Models (PLMs) being exploited, the research focus has gradually shifted towards PLM-based fine-tuning, known as the "pre-trained and fine-tuning" paradigm.

However, these detection approaches primarily belonged to the fully supervised learning paradigm, which heavily relies on constructing high-quality supervised datasets. The process of building such corpora is time-consuming and labor-intensive, leading to a challenging research bottleneck (Hu et al., 2022b; Shu et al., 2017). Additionally, despite the powerful ability of deep learning models to capture various news features, their transferability across domains is often limited, potentially leading to diminished performance in tasks involving domain shifts. In light of the scarcity of annotated data for detecting misinformation within specific domains, more researchers are drawn to conducting fake news detection with limited or no labeled data. The zero-shot/few-shot learning of fake news detection have emerged as a growing research trend, aiming to maintain competitive model generalizability under various real-life low-resource scenarios.

Investigation into social information dissemination reveals a interesting phenomenon driven by individuals' cognitive abilities and ideological biases: confirmation bias (Nickerson, 1998). As shown in Fig. 1, this suggests that social users, influenced by their own biases, tend to seek and accept information that confirms their beliefs, then share it. This forms collective opinions marked by cognitive bias, where the shared news tends to be consistently genuine or fake. Recently, researchers (Sharma et al., 2019; Shu et al., 2017) examined this phenomenon from psychological and sociological perspectives, highlighting factors like naive realism and confirmation bias, inherent in human nature. This biased behavior, fueled by personal beliefs, often ignores objective facts, leading to echo chambers and filter bubbles. We term this phenomenon as "news veracity dissemination consistency" and, in tandem with the previously discussed confirmation bias, we propose a theory known as the "user spreading news engagement bias". This suggests that social users who frequently repost the same news often reflect collective social opinions, influenced by a specific cognitive bias. Additionally, the authenticity of news shared by these collectives typically remains consistent, being mostly genuine or predominantly fake (Wu et al., 2023).

As shown in Fig. 2, calculated the relationships between reposted news by the same user across different news, where the dissemination frequencies exceeded 20 and 120 in the **PolitiFact**, **GossipCop** and **FANG** benchmarks (Shu et al., 2019b; Nguyen et al., 2020). It is evident to see that the left and right halves of Fig. 2 correspond to two distinct clusters of social context dissemination for real news and fake news, respectively. This confirms the "user spreading news engagement bias" theory, which is also applicable from the situation of news veracity dissemination, *i.e.*, news reposting veracity consistency.

Therefore, given the superiority of prompt-tuning in low-resource scenarios and multi-task learning in multi-source knowledge fusion, we propose PMTL-DisCo for fake news detection (FEND), a novel **P**rompting **M**ulti-**T**ask **L**earning framework guided by news veracity **Dis**semination **Co**nsistency reasoning augmentation.

The primary contribution of PMTL-DisCo, is to organically integrate the aforementioned "user spreading news engagement bias" theory with prompt-tuning paradigm through multi-task learning techniques to address the challenge of few-shot fake news detection in data-scarce scenarios.

The primary task of PMTL-DisCo is rumor verification, a binary classification task of whether textual news is a rumor, which is achieved through Mask Language Model (MLM) based pseudo prompt-tuning technique (Lee et al., 2021; Jin et al., 2024; Ma et al., 2022). This technique aims to leverage the pre-training objective of the pre-trained language models in bidirectional encoder representations from transformers (BERT) series, specifically the masked language modeling task, to re-formulate fake news detection as a masked token prediction problem. This technique effectively elicits task-specific knowledge from the Pre-trained Language Models (PLMs), significantly reducing the amount of supervised dataset it needs to learn (Wu et al., 2023).

In addition, we also introduced a novel expanded label words-based verbalizer engineering in prompt learning. The technique addresses the limitation in previous prompt-tuning where individual label word pairs (such as "real" and "fake", or "news" and "rumor") are insufficient to fully elicit the internal knowledge of PLM (Zhao et al., 2023). Specifically, the engineering primarily includes an improved acquisition approach for high-quality real/fake semantic-related label words (Wu et al., 2019a; Ma et al., 2022; Lenz and Bergmann, 2023; Jin et al., 2024; Jiao et al., 2020), as well as a expanded multi-label words mapping strategy based on dynamic attention mechanism and adaptive weight allocation (Hu et al., 2022a).

Building upon the "user spreading news engagement bias" theory, we constructed the Veracity Dissemination Consistency Network based

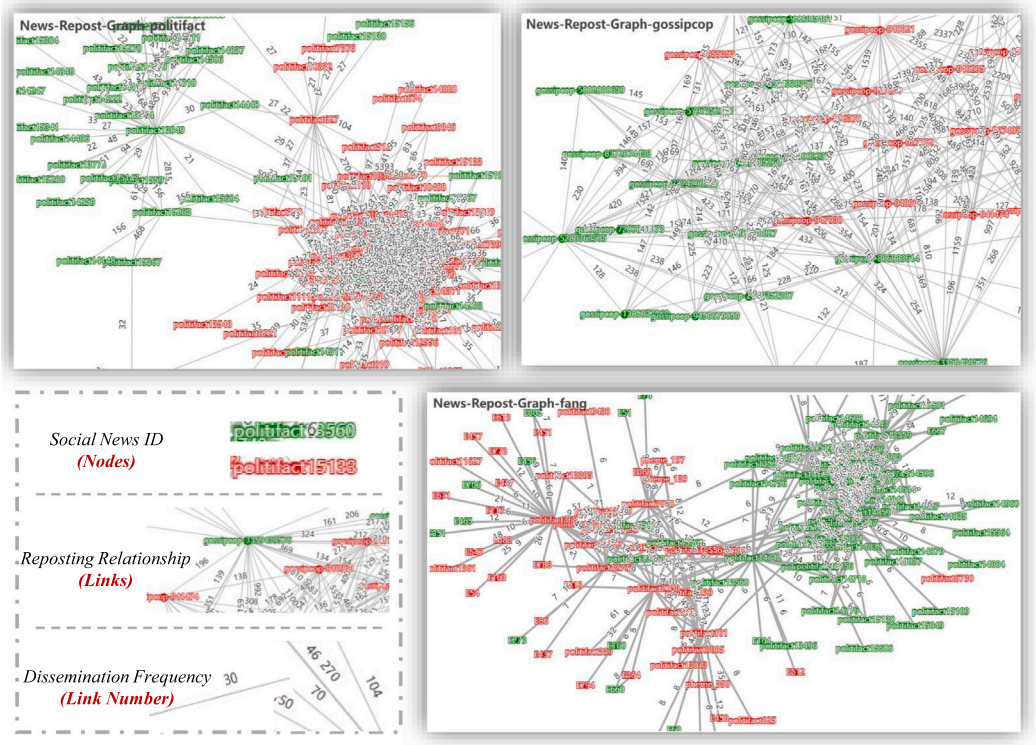


Fig. 2. A social context visualization of the news dissemination's authenticity consistency on the **PolitiFact** and **GossipCop** benchmarks (Shu et al., 2019b). The red and green nodes respectively represent real and fake news. The numbers on the links connecting the nodes indicate the dissemination frequency between two different pieces of news by the same social user.

on the collected social context-based news case dissemination records. The network is comprised of a multitude of real-world social news articles serving as nodes, with the connections between articles shared by the same social users depicted as links among the nodes. The network is utilized for the auxiliary task of "News Distributed Representation Optimization" in PMTL-DisCo. It also supports the "Multiple Neighboring News Reasoning Enhancement" strategy for the inference of PMTL-DisCo, facilitating the retrieval of first-order and second-order relationships among neighboring news.

To incorporate external supervisory signals regarding news veracity dissemination consistency, we integrated multi-task learning into our initial prompt learning framework Specifically, we utilize a auxiliary task called "news distributed representation optimization", in which the model is trained concurrently to optimize both the original rumor identification task objective and the auxiliary task objective. The "news distributed representation optimization" sub-task, utilizing the "user spreading news engagement bias" theory, helps facilitate the learning of semantic features of news under different social dissemination characteristics.

During the model inference, we employed a novel auxiliary strategy, namely "multiple neighboring news reasoning enhancement". This strategy, based on multiple neighboring news case reasoning, aims to optimize the predictions of a news case by utilizing the prediction results from news that have first or second-order social context dissemination relationships with the news case under prediction. It refines and integrates features between the prediction of current news case and the predictions of adjacent news.

Based on these improvements, even in scenarios with limited training instances, PMTL-DisCo can also exhibit remarkably competitive predictive capabilities.

Our contributions can be summarized as follows:

- We propose PMTL-DisCo, a prompting multi-task learning framework for few-shot fake news detection, which incorporates news veracity dissemination consistency reasoning into prompt-tuning through multi-task learning paradigm.
- PMTL-DisCo utilizes a auxiliary task, "news distributed representation optimization", to help model learn valuable semantic features between neighboring news under different social dissemination characteristics.
- We introduce a novel expanded label words-based verbalizer engineering in prompt-tuning. It consists of an improved acquisition approach for high-quality real/fake semantic-related label words, and an expanded multi-label words mapping strategy.
- We adopt a novel multiple neighboring news reasoning enhancement strategy, utilizing the predictions of multiple neighboring news case to optimize the prediction of the current news case, thereby improving the final FEND performance.

The remainder is organized as follows: Section 2 provides an overview of recent literature related to our work; Section 3 introduces experimental preliminaries, including task definition and several necessary notations; Section 4 describes our few-shot fake news detection framework: PMTL-DisCo; Section 5 presents the comprehensive experimental details, including the introduction of adopted datasets, metrics, and baselines, main comparisons (Section 5.2), and ablated studies (Section 5.3); Case study is detailed in Sections 5.3 and 6; Section 8 summarizes our primary contributions and future perspective for few-shot fake news detection.

2. Related works

2.1. Fake news detection (FEND)

With the widespread success of deep learning (DL) across various applications, DL-based FakE News Detection (FEND) methods have emerged and garnered considerable attention recently (Hu et al., 2022b; Sharma et al., 2019).

Nan et al. (2021) shifted their research focus to interdisciplinary detection of fake news. They firstly proposed a multi-expert framework based Multi-Domain Fake News Detection method (MDFEND),

Table 1
Comparative table of existing FEND methods and our PMTL-DisCo.

Method/ Researchers	Key features	Data requirements	Suitable for low-resource scenarios	Dynamic adaptation to news environment	Main limitations
MDFEND (Nan et al., 2021)	Multi-expert framework for cross-domain fake news detection.	Large-scale annotated datasets	No	No	Reliant on large-scale annotated data, struggles in few-shot scenarios.
NEP (Sheng et al., 2022)	News environment perception framework integrates macro and micro news environment influences	Large-scale annotated datasets	No	No	Requires comprehensive supervised learning framework; poor adaptability with few-shot data.
Prompt-and-Align (P&A) (Wu et al., 2023)	Few-shot fake news detection based on prompt-tuning and external knowledge from social context topology	Small amount of annotated data	**Yes**	No	Relies on static dissemination pattern representations; unable to dynamically adapt to changing news environments.
DECOR (Wu and Hooi, 2023)	Social graph optimization based on Graph Neural Networks using Degree-Corrected Stochastic Blockmodel.	Social graph data	No	No	Dependent on static social graphs; unable to dynamically adjust to different news environments.
Perplexity-based fact-checking (Lee et al., 2021)	Uses perplexity scores from evidence-conditioned language models without updating LM parameters.	Small amount of annotated data	**Yes**	No	Limited scalability due to reliance on optimal perplexity threshold.
VeraCT Scan (Niu et al., 2024)	Retrieval-augmented system extracts key factual claims.	Small amount of annotated data	**Yes**	No	High computational costs from utilizing large language models and extensive internet searches.
PMTL-DisCo (Ours)	Multi-task learning paradigm integrating an auxiliary task; Expanded label word mapping strategy; Multi-neighbor reasoning enhancement strategy.	Suitable for few-shot data	**Yes**	**Yes**	PMTL-DisCo performs better in low-resource scenarios, dynamically captures relationships between neighboring news, integrates dissemination patterns.

and publicly releasing the Chinese multi-domain fake news detection dataset "Weibo21". Sheng et al. (2022) broadened their focus beyond merely focusing on observing language pattern ("zoom in" mode), embracing a broader perspective that incorporates the crucial influence of the external news environment where fake news proliferates ("zoom out" mode). Recognizing the significance of both macro and micro news environment, they introduced the News Environment Perception Framework (NEP). Wu et al. (2023) proposed Prompt-and-Align (P&A), a novel method for few-shot fake news detection based on prompt-tuning. This approach simultaneously harnesses the pre-trained knowledge in PLMs and external knowledge from social context topology. They introduced a news proximity graph, represented by graph adjacency matrix, designed to capture the veracity-consistent signals in shared readerships. In the same year, Wu et al. also devised DECOR (Degree-Corrected Social Graph Refinement) (Wu and Hooi, 2023), a novel approach to enhancing fake news detection by refining noisy social graphs used in graph neural network (GNN). DECOR utilized degree-related patterns in social graphs—constructed based on user interactions with news articles—exhibit strong correlations with news veracity, employing a Degree-Corrected Stochastic Blockmodel (DCSBM) to iteratively refine edge weights in the graph, suppressing noise and improving the reliability of GNN-based veracity predictions. Lee et al. (2021) utilizes perplexity score from evidence-conditioned LMs to estimate the credibility of claims alongside their corresponding evidence. By identifying the optimal threshold to distinguish between "Supported" and "Unsupported" claims via evaluations on a validation set, this methodology achieves competitive and robust few-shot fact-checking capability without necessitating updates to any LM parameters. Niu et al. (2024) introduced VeraCT Scan framework, a retrieval-augmented system for fake news detection that extracts key factual claims from news articles and conducts internet-wide searches to validate these claims using corroborating or conflicting evidence. By leveraging advanced large language models like GPT-4 Turbo and Llama-2 13B, the system combines information retrieval, credibility assessment, and rationale generation to deliver interpretable and accurate veracity judgments.

2.2. Limitations of existing FEND methods

While existing Fake News Detection (FEND) methods have demonstrated significant advancements, they exhibit several limitations when applied to real-world scenarios, particularly in low-resource settings. For instance, methods like MDFEND (Nan et al., 2021) and NEP (Sheng et al., 2022) emphasize either multi-domain adaptability or the influence of macro and micro news environments but require large-scale annotated datasets to achieve optimal performance. This dependency on fully supervised learning frameworks limits their applicability in few-shot scenarios where labeled data is scarce. Graph-based approaches such as Prompt-and-Align (P&A) (Wu et al., 2023) and DECOR (Wu and Hooi, 2023) partially address the use of external context by leveraging social network structures. However, these methods rely heavily on static representations of dissemination patterns and fail to dynamically adapt to varying news contexts. Furthermore, as illustrated in Table 1, while prompt-tuning approaches such as P&A (Wu et al., 2023) and evidence-based frameworks like VeraCT Scan (Niu et al., 2024) show promise in few-shot settings, they do not incorporate multi-task learning paradigms or explicitly model dissemination consistency across socially connected news cases. Additionally, Lee et al. (2021) face limited scalability due to their reliance on optimal perplexity thresholds and the quality and adaptability of evidence-conditioned language models, and Niu et al. (2024) incur high computational costs from utilizing large language models and extensive internet searches, which can delay real-time detection.

In contrast, our proposed PMTL-DisCo framework introduces three key innovations to overcome these limitations: (1) A multi-task learning paradigm that integrates an auxiliary task, "news distributed representation optimization", to exploit dissemination consistency signals dynamically. (2) An expanded label words-based verbalizer engineering strategy that enhances prompt-tuning performance by addressing limitations in label word mapping. (3) A multi-neighbor reasoning enhancement strategy that leverages veracity features from socially connected

news, enabling more accurate predictions even in low-resource scenarios. These innovations collectively enable PMTL-DisCo to outperform existing FEND methods, as demonstrated through extensive experiments. By dynamically capturing the relationships between neighboring news and integrating them into the prediction process, PMTL-DisCo addresses a critical gap in existing research: the effective use of dissemination patterns in few-shot fake news detection.

2.3. News veracity dissemination consistency

Through investigation into the information dissemination, a cognitive abilities and ideological biases-driven social phenomenon has been identified: confirmation bias (Nickerson, 1998; Wu et al., 2023). This phenomenon suggests that social users, based on their own cognitive biases, tend to seek out and preferentially accept information that aligns with their existing beliefs, deeming it as correct, and then disseminate it.

Shu et al. (2017) analyzed the factors of fake news dissemination from a psychological perspective of confirmation bias. Sharma et al. (2019) examine and explain the existence and dissemination of fake news at both individual and societal levels towards the confirmation bias from the perspectives of sociology and psychology. This biased social behavior, motivated by personal cognition, positions, and beliefs, far surpasses verification against objective facts. These factors contribute to the formation of echo chambers and filter bubbles, thereby exacerbating the prevalence of confirmation bias.

The exacerbation of political and sentiment polarization in the United States is a typical example influenced by the fragmentation of news media and the dissemination of misleading information on social media (Kubin and von Sikorski, 2021). Researchers found that social users with similar ideologies in politics tend to cluster and engage more with allies, while distancing themselves from opposing forces. When this phenomenon is reflected in behaviors such as sharing real or fake political news, social users with consistent ideological viewpoints form collectives characterized by a significant amount of socially shared news dissemination with consistent information authenticity.

We refer to this phenomenon as the "news veracity dissemination consistency" and, in conjunction with the confirmation bias, derive a theory termed the "user spreading news engagement bias". Social users who repeatedly repost the same news often represent collective social opinions, characterized by a certain cognitive bias. Moreover, the authenticity of the news shared by these collectives tends to remain consistent, either mostly genuine or predominantly fake (Wu et al., 2023).

2.4. Prompt learning

Prompt learning relies on manually or automatically defined template prompts to transform natural language processing (NLP) tasks into text completion or text generation tasks. Recent research prove that it better harnesses and elicits the factual and commonsense knowledge acquired by the model during its pre-training stage (Liu et al., 2022; Zhao et al., 2023). Prompt learning can be categorized as:

- Autoregressive Language Model (LM)-based prompt-tuning based on unidirectional generative LM like GPT-series (Brown et al., 2020);
- Masked Language Model (LM)-based pseudo prompt-tuning based on bidirectional MLM like BERT-series (Devlin et al., 2019).

Autoregressive LM-based prompt learning. Autoregressive LM-based prompt learning integrates autoregressive LM's pre-training task, *i.e.*, self-regressive textual sequences generation. It crafts prompting templates based on specific task requirements and integrates them with original task inputs (Liu et al., 2022).

Li and Liang (2021) proposed Prefix-Tuning, a lightweight prompt-tuning method for various NLP generation tasks. This approach utilizes continuously trainable parameter matrices to adjust the feature representations of prompting templates. Zhao et al. (2023) introduced a prompt learning-based framework for metonymy resolution. Li et al. (2021) proposed SentiPrompt and pioneered the prompt-tuning for aspect-based sentiment analysis (ABSA) by using BART. Cui et al. (2021) proposed BART-based TemplateNER, which is the first prompt-tuning application to the sequence labeling task of named entity recognition (NER). Hu et al. (2022a) developed knowledgeable prompt-tuning (KPT), an innovative approach leveraging external knowledge bases to enhance the knowledge of verbalizers. This mitigates biases introduced by personal vocabulary while ensuring sufficient coverage.

Masked LM-based pseudo prompt learning. Mask Language Model (MLM) based pseudo prompt learning incorporates Mask Language Model's original per-training task, including BERT, RoBERTa and so on Wu et al. (2019a) and Ma et al. (2022).

Lee et al. (2021) utilized the pseudo perplexity generated from Masked LM to solve the few-shot fact-checking problem. Jin et al. (2024) introduced WordTransABSA to address the aspect-based sentiment analysis task, where they made slight modifications to the Masked LM pre-training task by introducing a novel "word transferred LM" fine-tuning approach. Ma et al. (2022) proposed a template-free prompt-tuning approach for few-shot NER. It adapts the Mask Language Model (MLM) pre-training task of BERT to an enhanced optimization task focused on Entity-Oriented Language Modeling.

Overall, the remarkable few-shot learning scenarios capability of prompt learning inspires us. The PMTL-DisCo proposed in this work also leverages Masked LM-based Pseudo prompt learning.

2.5. Multi-task learning

The multi-task learning enables the transfer of essential inductive bias (which can be viewed as external guiding signals or knowledge) from auxiliary tasks to the target task (Zhang and Yang, 2022; Yang et al., 2024; Zheng et al., 2024). Multi-task learning is a machine learning framework designed to extract valuable insights from various interconnected sub-tasks, thereby boosting the overall performance and generalization ability across all tasks, or employing supplementary tasks to support the primary task objective.

Wu et al. (2019b) introduces a sifted multi-task learning method with a selected sharing layer for fake news detection, utilizing gate and attention mechanisms to filter and select shared feature flows between tasks, resulting in state-of-the-art performance improvements on two widely used competition datasets. Choudhry et al. (2022) proposed a multi-task framework for fake news and rumor detection, incorporating emotion class labels to enhance model feature extraction performance. Experimental results demonstrated the framework consistently outperforms single-task models in both within-domain and cross-domain settings. Zhou et al. (2019) developed a parameter-sharing-based multi-task learning framework for Relation Extraction (MTRE), employing a hierarchical neural network. This model integrates dependency and entity representations from two auxiliary tasks: entity type classification and dependency parsing.

Inspired by these, our work incorporates the parameter-sharing-based multi-task learning technique, leveraging the sub-task, news distributed representation optimization to provide helpful inter-task transferred bias, refining and optimizing the primary task, prompt learning-based rumor verification.

3. Methodology preliminaries

3.1. Notations

For the fake news detection (FEND) task, each news to be detected along with its predicted veracity category can be regarded as a independent news case. Consider a large-scale network composed of numerous

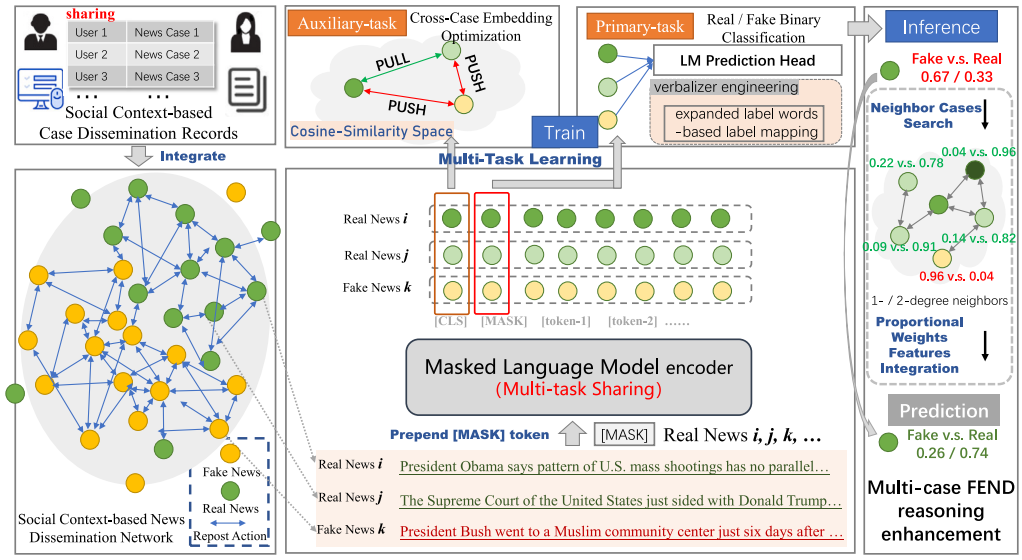


Fig. 3. The model architecture of our proposed PMTL-DisCo, which is a prompting multi-task learning framework. Utilizing the constructed veracity dissemination consistency network (left) to train the "news distributed representation optimization" task; Leveraging the Masked LM-based pseudo prompt-tuning to conduct the primary task, Real/Fake binary classification for FEND (center); And the adopted multiple neighboring news reasoning enhancement strategy during the FEND inference (right).

news, where different news directly connected are related to the same user's dissemination. We refer to it as the "Social Context-based News Dissemination Network".

The enhancements based on multiple neighboring news reasoning is the significant contribution of our proposed PMTL-DisCo. Specifically, leveraging the aforementioned network, we devised two major dissemination consistency-guided reasoning augmentation techniques during both the multi-task training and the model inference procedure:

1. During the multi-task learning, we utilize a "news distributed representation optimization" auxiliary task to enable the model to learn relationship features between different news in the veracity dissemination consistency network, which can be regarded as external enhancement signals for fake news detection;

2. During the inference/detection process, the "multiple neighboring news reasoning enhancement" strategy we proposed involves utilizing the reasoning results of news in the veracity dissemination consistency network, which are first- and second-degree neighbors to the news case to be detected, in order to optimize the original detection results of the news case.

3.2. Task definition

Consider a few-shot dataset $\mathcal{D} = [\mathcal{T}, \mathcal{V}] = [\{s, l\}_{i=1}^{N}; \{s, l\}_{j=1}^{M}]$ for Fake News Detection (FEND), comprising a train set $\mathcal{T} = \{s, l\}_{i=1}^{N}$ with N training samples and a test set $\mathcal{V} = \{s, l\}_{j=1}^{M}$ with M testing samples, where $N \ll$ M. Each sample contains a set of a news article s along with its corresponding authenticity label l. Alongside the aforementioned FEND dataset $\mathcal{D}$, we incorporate a set of social context-based news case dissemination records $\mathcal{R}$ representing the user engagements of news reposting. The notation $\mathcal{R}$ is formulated as $\{(u, d) \mid u \in U, d \in \mathcal{D}\}$ where U is the user IDs collection involved in reposting news from $\mathcal{D}$, denoting user u's engagement in disseminating news from $\mathcal{D}$.

Given the above defined notations and prerequisites, the FEND task can be formulated as:

Given: a few-shot dataset $\mathcal{D}' = [\mathcal{T}, \mathcal{V}'] = [\{s, l\}_{i=1}^{N}; \{s\}_{j=1}^{M}]$ without the labels of train set $\mathcal{V} - \mathcal{V}'$, and a social context-based news dissemination records set $\mathcal{R}$;

Target: predicting the correct labels $\{l\}_{j=1}^{M}$ for test set $\{s\}_{j=1}^{M}$ from $\mathcal{D}$.

4. Our approach (PMTL-DisCo)

In this section, we provide an overview of our proposed **PMTL-DisCo** for fake news detection: a **P**rompting **M**ulti-**T**ask **L**earning framework guided by News Veracity **Dis**semination **Co**nsistency Reasoning Augmentation. We will also illustrate the detailed operational principles the major components in PMTL-DisCo. Given that the PMTL-DisCo relies on various enhancement strategies, we will delineate the workflow of each component in separate subsections, including:

Section 4.1: The construction steps of the Veracity Dissemination Consistency Network;

Section 4.2: The training process for the primary task of "prompt learning-based rumor verification" (*i.e.*, real/fake label binary classification for FEND) by utilizing the initial few-shot train set $\mathcal{T} = \{s, l\}_{i=1}^{N}$ and the Masked LM-based pseudo prompt-tuning technique;

Section 4.3: The implementation details of the "expanded label words-based verbalization engineering" introduced in MLM pseudo prompt-tuning, including specific implementation processes for acquiring high-quality label words and an adaptive weight allocation-based multi-label mapping strategy;

Section 4.4: The details of "news distributed representation optimization" auxiliary task based on the aforementioned Veracity Dissemination Consistency Network and Masked Language Model encoder (e.g. BERT encoder, RoBERTa encoder);

Section 4.5: The computation of joint loss objective in multi-task learning, along with its optimization training characteristic for the sub-tasks described above.

Section 4.6: The specific implementation of the "multiple neighboring news reasoning enhancement" strategy during the inference/detection of PMTL-DisCo.

4.1. Construction of veracity dissemination consistency network

In Section 1, Introduction, we focused on analyzing the social phenomenon: "confirmation bias", and the characteristics brought by user behavioral preferences to the fake news dissemination: "news veracity dissemination consistency". We elaborated on the corresponding theory "user spreading news engagement bias" and its positive meaning to the FEND task, serving as effective external knowledge for rumor verification in few-shot scenarios, providing additional external cross-case signals.

Building upon the aforementioned "user spreading news engagement bias" theory, we constructed a undirected graph in this section, known as Veracity Dissemination Consistency Network. The network is comprised of a multitude of real-world social news articles serving as nodes, with the connections between articles shared by the same social users depicted as links among the nodes. It is utilized for the auxiliary task of "News Distributed Representation Optimization" in the PMTL-DisCo, aiming to directly learn the space distributional characteristics of the dissemination consistency network. Additionally, it supports the "Multiple Neighboring News Reasoning Enhancement" strategy for the PMTL-DisCo inference, streamlining the retrieval of first- and second-order neighboring news from the extensive pool of cases.

Algorithm 1 provides a detailed description of the construction process of the Veracity Dissemination Consistency Network. This network is built using the FEND dataset $\mathcal{D}$ and the social context-based case dissemination records $\mathcal{R}$. As described in this algorithm, the construction involves a series of data processing steps. The algorithm generates the undirected graph $\mathbf{G} = (\mathcal{N}, \mathcal{L})$, Veracity Dissemination Consistency Network. It consists of Nodes $\mathcal{N}$ representing news article cases and Links $\mathcal{L}$ denoting shared-reposting relationships among news. $\mathbf{G}$ vividly illustrates the interconnections between these news, showcasing the level of sharing through the frequency of reposting by the same social user.

For efficient retrieval of massive graph nodes and edges in large-scale deployments on this veracity consistency networks, we recommend utilizing cutting-edge graph databases such as **Neo4j**[2] and **Nebula Graph**.[3] In this experiment, we opted for **Neo4j**, an open-source tool known for its advanced storage technology, to ensure optimal read and write efficiency.

Algorithm 1 The Construction Steps of Veracity Dissemination Consistency Network.

Input: Social Users U; News $\mathcal{C} = [\mathcal{T}; \mathcal{V}] = [\{s\}_{i=1}^{N}; \{s\}_{j=1}^{M}]$ from both train set $\mathcal{T}$ and test set $\mathcal{V}$ in FEND dataset $\mathcal{D}$; Social context-based case dissemination records $\mathcal{R} = \{(u, c) | u \in U, c \in \mathcal{C}\}$;

Output: Veracity Dissemination Consistency Network $\mathbf{G} = (\mathcal{N}, \mathcal{L})$, where $\mathcal{N}$: Nodes representing news article cases shared and disseminated by social users; and $\mathcal{L}$: Links among nodes $\mathcal{N}$, indicating connections/relationships between news shared by the same social user.

Initialize: $\mathcal{N} \longleftarrow []$; $\mathcal{L} \longleftarrow []$;
Initialize: Threshold **M** of News Reposting Frequency for Active Social Users;
Initialize: Threshold **N** of the News Reposting Frequency for Hotspot News;
% Filter and Retain Active Social Users
% Filter and Retain Hotspot News
$\mathcal{U}' \longleftarrow []$; $Count_{\mathcal{U}'} \longleftarrow \{\}$;
$\mathcal{C}' \longleftarrow []$; $Count_{\mathcal{C}'} \longleftarrow \{\}$;
for all r in $\mathcal{R}$ **do**
  $(u, c) \leftarrow r$;
  Update $Count_{\mathcal{U}'}$ with u and $Count_{\mathcal{U}'}[u] += 1$;
  Update $Count_{\mathcal{C}'}$ with c and $Count_{\mathcal{C}'}[c] += 1$;
end for
for all u, c in $Count_{\mathcal{U}'}$, $Count_{\mathcal{C}'}$ **do**
  if $Count_{\mathcal{U}'}[u] >$ **M then**
    Add u to $\mathcal{U}'$;
  end if
  if $Count_{\mathcal{C}'}[c] >$ **N then**
    Add c to $\mathcal{C}'$;
  end if
end for
Obtained: $\mathcal{U}', \mathcal{C}'$
% Veracity Dissemination Consistency Network Construction
$\mathcal{N} \longleftarrow []$; $\mathcal{L} \longleftarrow \{\}$;
for all $\hat{r}$ in $\mathcal{R}$ **do**
  $(\hat{u}, \hat{c}) \leftarrow \hat{r}$;
  if $\hat{u} \in \mathcal{U}', \hat{c} \in \mathcal{C}'$ **then**
    for all $\overline{r}$ in $\mathcal{R}$ **do**
      $(\overline{u}, \overline{c}) \leftarrow \overline{r}$;
      if $\hat{u} == \overline{u}$ && $\hat{c}$!= $\overline{c}$ **then**
        if $\hat{u}$ not in $\mathcal{N}$ **then**
          Add $\hat{u}$ to $\mathcal{N}$;
        end if
        if $(\hat{c}, \overline{c})$ not in $\mathcal{L}$ **then**
          Update $\mathcal{L}$ with $(\hat{c}, \overline{c})$ and $\mathcal{L}[(\hat{c}, \overline{c})] = 0$;
        else
          $\mathcal{L}[(\hat{c}, \overline{c})] += 1$;
        end if
      end if
    end for
  end if
end for
% Delete duplicated graph links from $\mathcal{L}$
for all i in $\mathcal{L}$; j in $\mathcal{L}$ **do**
  $(\hat{c}_i, \overline{c}_i) \leftarrow i; (\hat{c}_j, \overline{c}_j) \leftarrow j$
  If: $\hat{c}_i == \overline{c}_j$ && $\overline{c}_i == \hat{c}_j$
  Then: Delete item $(\hat{c}_j, \overline{c}_j)$ from $\mathcal{L}$;
end for
Save $(\mathcal{N}, \mathcal{L})$ as Undirected Graph **G** to Neo4j or Nebula Graph;
Return: Undirected Graph **G**, *i.e.*, $(\mathcal{N}, \mathcal{L})$.

4.2. Prompt learning-based rumor verification

The "prompt learning-based rumor verification" is the primary task that used for FEND (real/fake label binary classification). In contrast to Autoregressive LM-based prompt-tuning, we adopt Mask Language Model (MLM) based pseudo prompt-tuning in the PMTL-DisCo framework. Specifically, this fine-tuning technique is performed on Mask Language Models such as BERT, RoBERTa, ELECTRA, and SpanBERT (Devlin et al., 2019; Wu et al., 2023). Compared to the unidirectional contextual semantics of Autoregressive LMs, the backbone encoder of pseudo prompt-tuning: Mask Language Model, can encompass contextual information and provide a richer representation of global textual semantics for news.

We employed the notation $\mathcal{PLM}$ to denote the Mask Language Model utilized in PMTL-DisCo. In our experiment, $\mathcal{PLM}$ specifically refers to "BertForMaskedLM" of "bert-base-uncased".[4] BertForMaskedLM not only incorporates the original BERT encoder but also integrates a decoder, which is a multi-layer perceptron network with a vocabulary-sized output dimension, reused from BERT in the MLM pre-training task. Given an sentence input, BertForMaskedLM provides the probability distribution of vocabulary words for each token position in the original text, thus faithfully simulating the MLM pre-training task during its pre-training stage.

Firstly, given a news article $\mathbf{S} = \{s_i\}_{i=1}^{N}$ composed of N words ($\mathbf{S} \in C$), we manually construct a prompt template containing special [MASK] tokens and slot of news article **S**, which is successive served as the prompt fed to PLM.

For the sake of simplicity, in our experiment, we utilize a single also simple fixed template **T**: "It is a [MASK] news: [SLOT of **S**]". By

[2] Neo4j: https://neo4j.com/ *[Accessed on 2024.03]*.
[3] Nebula Graph: https://www.nebula-graph.com.cn/ *[Accessed on 2023.03]*.
[4] Google BERT: https://huggingface.co/google-bert/bert-base-uncased *[Accessed on 2023.02]*.

integrating the news article **S** with the prompt template **T**, we obtain the PLM input $\hat{\mathbf{S}}$, which can be formulated as Eq. (1):

$$\hat{\mathbf{S}} = \mathbf{T}(\mathbf{S}) = It\ \ is\ \ a\ \ [MASK]\ \ news:\ \ \mathbf{S}. \tag{1}$$

$$\mathcal{V} = \mathcal{PLM}(\hat{\mathbf{S}}) = [v_{[CLS]}, v_{It}, \ldots, v_{[MASK]}, \ldots, v_{[SEP]}]. \tag{2}$$

As depicted in Eq. (2), taking $\hat{\mathbf{S}}$ as $\mathcal{PLM}$ input, BertForMaskedLM outputs a probability distribution of vocabulary words for each token in the textual input, represented by the sequence $\mathbb{V}$. Here, $v_i \in \mathbb{R}^{|\mathcal{V}|}$, where $\mathcal{V}$ denotes the length of the PLM vocabulary. We consider optimizing the vocabulary probability distribution $v_{[MASK]}$ at the token position of [MASK].

In the inference step, the logits vector $v_{[MASK]}$ representing the probability distribution of the vocabulary at [MASK] position is provided, the subsequent PMTL-DisCo decoding process is closely associated with the mainstream Answer Mapping Engineering in prompt learning (Liu et al., 2022). In previous prompt-based learning tasks for multi-classification, such as those involving N categories denoted as $\mathbb{C} = c_1^N$, the initial step predominantly revolves around establishing a series of label word sets $\mathbb{L} = \mathcal{L}_1^N$ for every pre-defined N categories, where each label word set $\mathcal{L}$ corresponds to an independent category c. These task involves determining the real category $\hat{c}$ by identifying its corresponding label word set $\hat{\mathcal{L}}$ based on its label word l that maximizes the probability at the [MASK] position.

Given the logits represented as $v_{[MASK]}$, and $P(i, j)$ indicating the function to determine the probability of word j from logits vector i, along with the function $Map(\cdot)$ which maps from $\hat{\mathcal{L}}$ to $\hat{c}$, *i.e.*, $\mathbb{L}$ to $\mathbb{C}$, this reasoning process can be elucidated as follows:

$$\hat{c} = Map(\text{Argmax}_{l \in \hat{\mathcal{L}}} P(v_{[MASK]}, l), \mathbb{L}). \tag{3}$$

This process involves an "Argmax" search function to find the label word l with the highest probability score derived from the [MASK] token position's probability distribution $v_{[MASK]}$. Subsequently, this label word is mapped back to the pre-defined categories for the target task.

However, the adoptions of single label word and simple category mapping strategy require significant prior experimentation. The randomness associated with a single label word can greatly affect the effectiveness of the answer search process, making the prompt-tuning performance susceptible to the suitability of the chosen label word. Therefore, in our PMTL-DisCo, we propose a novel expanded label words-based verbalizer engineering. For details, please refer to Section 4.3.

Overall, the MLM-based pseudo prompt-tuning optimizes the primary task "prompt learning-based rumor verification" of PMTL-DisCo by fine-tuning the words probability distribution at the '[MASK]' positions to predict the final binary classification result of fake/real. It fundamentally differs from traditional fine-tuning, where a simple target label classification network is attached to the vector representation outputted by $\mathcal{PLM}$ encoder. MLM-based pseudo prompt-tuning fully exploits the rich linguistic knowledge embedded in $\mathcal{PLM}$, resulting in competitive performance compared to traditional fine-tuning methods (Jin et al., 2024).

4.3. Expanded label words-based verbalization engineering

As stated above (Section 4.2), the used label words are singular in mainstream verbalizer engineering, insufficient to cover the category features of the downstream linguistic processing tasks. For example, in Prompt-Align (Wu et al., 2023), a recently proposed rumor detection method, only two label words corresponding to the real/fake categories are defined: "news", "rumor". Prompt-Align optimizes the probability distribution of words at [MASK] position and maps "news"→ 0 (real) and "rumor"→ 1 (fake) to the target categories. Such strategies rely on the quality of the selected label words, leading to poor model generalization and robustness.

To address these challenges, we propose a novel expanded label words-based verbalizer engineering. This approach consists of two strategies:

- 1. An improved approach for acquiring high-quality real/fake semantic-related label words;
- 2. An expanded multi-label words mapping strategy based on dynamic attention mechanism and adaptive weight allocation.

The two strategies are respectively responsible for:

- 1. Searching and selecting enough high-standard expanded label words from external knowledge bases;
- 2. Using the expanded multi-label words to enhance the words mapping procedure in verbalization engineering during PMTL-DisCo optimization and prediction.

4.3.1. Expanded label words acquisition

As shown in Fig. 4, we employed various tools and techniques to search through external knowledge repositories, including WordNet, ChatGPT, Word Embedding, Masked LM, to filter high-quality label words (Lenz and Bergmann, 2023; Ma et al., 2022; Jin et al., 2024). In our experiments, we adopted three approaches to obtain label words for the fake and real categories:

- **WordNet SynSet Substitution**: This technique involved manual retrieval through WordNet, a thesaurus that offers synonyms for any given word[5];
- **Word Embedding Substitution**: We employed the Word Embedding technology to find words with semantic similarity as synonyms (Chaudhary, 2020);
- **MLM-based Masked Word Substitution**: Using sentences generated by ChatGPT containing real or fake content, we calculated the probability distribution of masked words at "real" and "fake" positions through Masked LM. We then selected high-frequency words with high probabilities that could replace the original masked words.

During the retrieval of WordNet, we employed a subset of SynSets from WordNet, which are interconnected through conceptual semantics and lexical relations. WordNet is a large English lexical database, manually curated, containing relationships between words, and it serves as a valuable tool in computational linguistics and natural language processing. Nouns, verbs, adjectives, and adverbs are grouped into sets of cognitive synonyms (synsets), with each synset representing a unique concept.

Specifically, we begin with the "real" and "fake" words as our initial label words, and then we retrieve their corresponding sets of synonyms, known as SynSets, from WordNet. There are a total of 117,000 such SynSets in WordNet (Lenz and Bergmann, 2023; Jin et al., 2024). Each SynSet is connected to other synsets through conceptual relations. In our experiments, to ensure comprehensive retrieval results, we not only consider the words within a single synset but also include words from synsets that have hypernym or hyponym relations with them. To balance coverage and complexity, we only consider synsets connected through first-order and second-order conceptual relations.

Simultaneously, we use ChatGPT[6] to generate expanded label words that enrich the representation of "real" and "fake" news. This approach refines the label mapping process, ensuring that the predictions are more interpretable and accurate. The generated label words are subsequently integrated into the prompt-tuning process for few-shot rumor verification. To ensure reproducibility and address potential concerns

[5] Princeton WordNet: https://wordnet.princeton.edu/ *[Accessed on 2024.02]*.

[6] You can access ChatGPT by visiting the following URL: https://chat.openai.com/chat [Accessed on 2023.12].

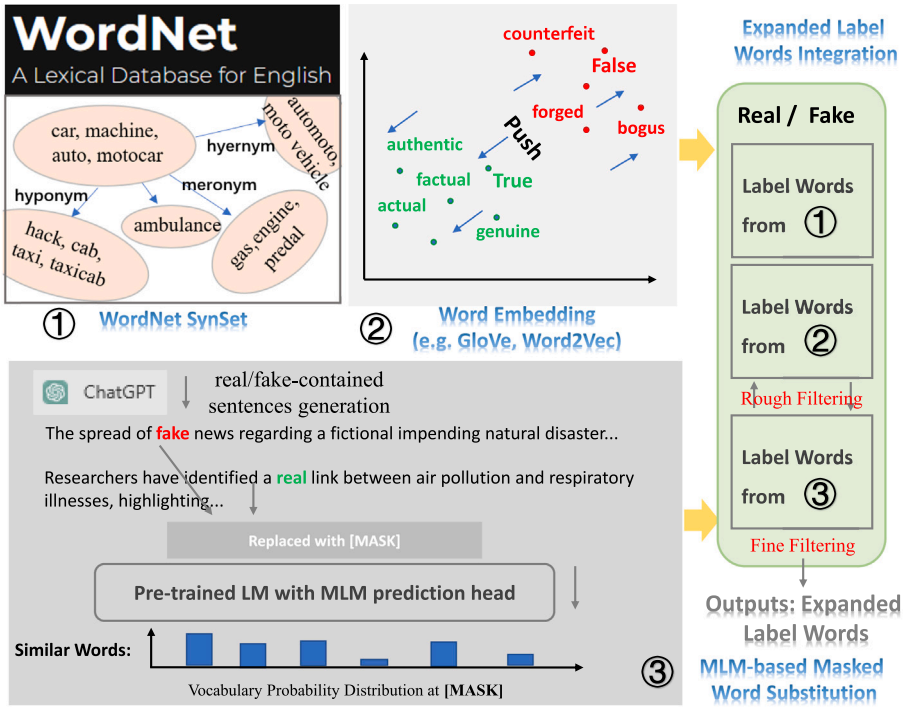


Fig. 4. The illustration of our proposed novel approach for acquiring high-quality real/fake semantic-related label words.

Table 2
Configuration details for GPT in label word acquisition.

Parameter	Value	Description
Model version	GPT-3.5-turbo	(released December 2023)
Temperature	0.7	(balances creativity and coherence)
Top-p (nucleus sampling)	0.9	(filters high-probability words and phrases)
Max tokens	50	(ensures concise and relevant label word generation)
Prompt structure	Custom-designed for fake news detection with task-specific context and examples	"You are an advanced language model tasked with assisting in the generation of high-quality label words for a fake news detection system. Your goal is to identify and suggest synonyms or semantically related words for the categories 'fake' and 'real', which will serve as label words for a classification task. No less than 50 words".

regarding the variability in GPT-generated outputs, we provide details about the ChatGPT version and settings used. We employed OpenAI's `GPT-3.5-turbo` model (December 2023 version) for generating label words.

The settings shown in Table 2 were applied during label word generation. These settings were consistent across all experiments to minimize variability. This ensures that future researchers can replicate our work using the same label words or adjust their experiments accordingly.

Similarly, during the Word Embedding Substitution, we adhere to the definition of "real" and "fake" words as our initial label words. Subsequently, we employ both Word2Vec and GloVe techniques to seek words semantically akin to the initial label words. In Word2Vec and GloVe, the Euclidean distance (or cosine similarity) between two word vectors serves as an effective metric for gauging the linguistic or semantic affinity among corresponding words. Occasionally, the nearest neighbors based on this metric unveil uncommon yet relevant words that lie outside an average human lexicon. Following this, we leverage the Python toolkit, specifically within the Gensim class `downloader.api` and its `most_similar` function, to calculate the *Top* similar words closest to the initial label words. Finally, we take the intersection of the outputs from Word2Vec and GloVe, constituting the retrieval results of WordNet SynSet Substitution.

In addition to the aforementioned techniques for label word retrieval, we can also expand label words through methods such as back translation, introducing random spelling errors, and leveraging generative language models (Chaudhary, 2020). These efforts will be left for future exploration and optimization.

Given that our PMTL-DisCo leverages MLM-based pseudo prompt-tuning, especially in fine-tuning on few-shot data, the selected high-quality label words should preferably align with the LM's probability preferences. Therefore, as illustrated in Fig. 4 right, during label words post-processing, we first applied coarse filtering to the label words using channels 1 (WordNet SynSet Substitution) and 2 (Word Embedding Substitution). Then, we conducted further refinement (fine filtering) using channel 3 (MLM-based Masked Word Substitution) to achieve finer selection.

Based on the WordNet SynSet and Word Embedding Substitution channels, the sets of synonymous real and fake words were obtained, denoted as $\mathcal{W}_{syn} = \{\mathcal{W}_{syn}^{real}; \mathcal{W}_{syn}^{fake}\}$ and $\mathcal{W}_{embed} = \{\mathcal{W}_{embed}^{real}; \mathcal{W}_{embed}^{fake}\}$, respectively. Along with the MLM-based Masked Word Substitution, we further filter out the final label words used for the expanded multi-label words mapping strategy based on the results of the previous two channels. The procedures of the MLM-based Masked Word Substitution and the post-processing steps are outlined in Algorithm 2.

This algorithm elaborates on the procedure of MLM-based Masked Word Substitution. During the label words post-processing, we remove words that are not in $\mathcal{W}_{syn}$ and $\mathcal{W}_{embed}$. After coarse filtering, we remove words that are appeared in both D_{real} and D_{fake}. Finally, for each

Algorithm 2 The steps of the MLM-based Masked Word Substitution and the label words post-processing.

Input: $\mathcal{W}_{syn}$, synonymous sets of real/fake from WordNet SynSet;
$\mathcal{W}_{embed}$, synonymous sets of real/fake from Word Embedding;
ChatGPT for real/fake-contained sentences generation;
BertForMaskedLM for masked word prediction;
Output: Final expanded label words $\hat{\mathcal{W}}$;

Initialize: Num of Generated Sentence, $\mathcal{K}$; Num of MLM predicted candidate words: $\mathcal{M}$; Num of each category's label words: $\mathcal{N}$;
$S_{real} = \{r_i, ..., r_{\mathcal{K}}\}$, and $S_{fake} = \{f_i, ..., f_{\mathcal{K}}\}$ from ChatGPT with artificial prompts;
% `MLM-based Masked Word Substitution`
Initialize BertForMaskedLM model using the official pre-trained weights, denoted as $\mathcal{PLM}$; $D_{real} \longleftarrow \{\}, D_{fake} \longleftarrow \{\}$;
for all $r_i \in S_{real}$, $f_i \in S_{fake}$ **do**
  Replace the positions of "real" and "fake" with [MASK], getting $\bar{r}_i$, $\bar{f}_i$;
  $T_i^{real} \longleftarrow \mathcal{M}$ most probable words from $\mathcal{PLM}$ based on $\bar{r}_i$;
  $T_i^{fake} \longleftarrow \mathcal{M}$ most probable words from $\mathcal{PLM}$ based on $\bar{f}_i$;
  for all x $\in T_i^{real}$, y $\in T_i^{fake}$ **do**
    if x/y in D_{real}/D_{fake} **then**
      $D_{real/fake}[x/y] += 1$;
    else
      $D_{real/fake}[x/y] = 0$;
    end if
  end for
end for
% `Coarse filtering`
Delete words in D_{real} and D_{fake} that does not exist in $\mathcal{W}_{syn}$ and $\mathcal{W}_{embed}$;
% `Fine filtering`
Remove words that appear simultaneously in both the D_{real} and D_{fake};
Arrange words in D_{real} and D_{fake} based on their occurrence frequencies;
Remove words from D_{real} and D_{fake} sets that are not sorted within $\mathcal{N}$;
Return: Final expanded label words $\hat{\mathcal{W}} = D_{real}$ and D_{fake}.

real/fake category, we retain the top **N** words as the ultimate expanded label words. In the experiment, we chose the hyperparameter N to be 10, which simultaneously ensures both comprehensive semantic coverage and minimal complexity, keeping computational and cognitive costs low.

4.3.2. Expanded multi-label words mapping strategy

As described in Section 4.2, the logits vector $v_{[MASK]}$ representing the vocabulary probability distribution at the [MASK] position is the prediction of the Masked Language Model for the word most likely to appear at that position.

Traditional prompt-tuning methods for integrating label words typically involve weighted averaging of labels or selecting the label word with the highest probability for each category, to map them into the target categories. However, in few-shot learning scenarios, these methods are susceptible to noise, resulting in significant fluctuations in model prediction performance.

Based on the collected expanded label word sets $\mathcal{L}_{real}$ and $\mathcal{L}_{fake}$ for the categories C_{real} and C_{fake}, we devised an expanded multi-label words mapping strategy during PMTL-DisCo optimization and prediction, utilizing dynamic attention mechanism and adaptive weight allocation. We employ dynamic attention mechanism and adaptive weight allocation to provide a more stable label word mapping approach.

Specifically, for each category, such as C_{fake}, and for the ith label word l_i^{fake} in $\mathcal{L}_{fake}$, we assign a learnable weight w_i^{fake} to it. These weights of w_i^{fake} form a weighting vector $\mathbf{w}^{fake}$ for the C_{fake} category, $\mathbf{w}^{fake} = [..., w_i^{fake}, ...]$, where $\mathbf{w}^{fake} \in \mathbb{R}^{|\mathcal{L}_{fake}|}$.

Specifically, in our experiments, these weights are derived by passing through multiple layers of MLP and activation functions after the logits vector $v_{[MASK]}$, and finally applying SoftMax operation. Notably, the dimensionality of the last MLP layer equals the words count $|\mathcal{L}_j|$ of the label words set $\mathcal{L}_j$ for each category C_j.

The weights $\mathbf{w}^j$ within each category C_j's label word set $\mathcal{L}_j$ are then normalized into $\alpha_{l_i^j}$ as follows:

$$\alpha_{l_i^j} = \frac{\exp\left(w_i^j\right)}{\sum_{l_k^j \in \mathcal{L}_j, w_k^j \in \mathbf{w}_j} \exp\left(w_k^j\right)}. \tag{4}$$

where $\alpha_{l_i^j}$ represents the normalized adaptive weight of the ith label word l_i^j in category C_j; w_i^j and w_k^j respectively refer to the allocated weights of l_i^j and l_k^j before weighting normalization.

Utilizing the computed normalized adaptive weights for label words, we aggregate the weighted probabilities of all label words in each category. This aggregation involves calculating a weighted sum, and based on the resulting total weight for each category, we ascertain the ultimate category label for the FEND task.

Intuitively, throughout the training process, we anticipate that smaller weights will be learned for noisy label words, thereby minimizing their impact on predictions.

Based on this expanded multi-label words mapping strategy, the optimization objective of the prompt-tuning rumor verification task is designed to maximize the sum of the adaptive weighted allocated probabilities of the label words under the correct label set.

Given the original input S and its gold category $\widetilde{C}$, "BertForMaskedLM" outputs the vocabulary probability distribution $v_{[MASK]}$ at the position of [MASK] token, $v_{[MASK]} = [p_{t_0}, \ldots, p_{t_i}, \ldots, p_{t_{|V|-1}}]$, in which $v_{[MASK]}$ comprises of the probability p_{t_i} of each words t_i in the PLM vocabulary V, along with the normalized adaptive weights for all label words $\boldsymbol{\alpha} = [..., \alpha_{l_i^j}, ...]$, in which $\alpha_{l_i^j}$ corresponds to the ith label word l_i^j of category C_j (See Eq. (4)), the optimization objective is formulated as follows.

$$Maximize\, f(v_{[MASK]}, \boldsymbol{\alpha}) = \sum_{l_i^{\widetilde{C}} \in \mathcal{L}_{\widetilde{C}}} p_{l_i^{\widetilde{C}}} \cdot \alpha_{l_i^{\widetilde{C}}}. \tag{5}$$

where the $\mathcal{L}_{\widetilde{C}}$ denotes the label words set corresponding to the gold category $\widetilde{C}$; .

Hence, the loss function L_{rv} of rumor verification utilized the standard binary cross-entropy loss $CE(\hat{y}, y') = -\hat{y}\,log(y') - (1-\hat{y})\,log(1-y')$, and thus is formulated as follows.

$$L_{rv} = -\widetilde{C} \cdot log\Big(\sum_{l_i^{\widetilde{C}} \in \mathcal{L}_{\widetilde{C}}} p_{l_i^{\widetilde{C}}} \cdot \alpha_{l_i^{\widetilde{C}}}\Big) - (1-\widetilde{C}) \cdot log\Big(\sum_{l_i^{\neg\widehat{C}} \in \neg\mathcal{L}_{\widehat{C}}} p_{l_i^{\neg\widehat{C}}} \cdot \alpha_{l_i^{\neg\widehat{C}}}\Big). \tag{6}$$

where the $\neg\widehat{C}$ denotes the erroneous category contrasting with the gold category $\widehat{C}$; the $\neg\mathcal{L}_{\widehat{C}}$ denotes the corresponding label words set of the erroneous category $\neg\widehat{C}$; the $p_{l_i^{\neg\widehat{C}}}$ and $\alpha_{l_i^{\neg\widehat{C}}}$ represent the ith label word probability and its normalized adaptive weight of the erroneous category $\neg\widehat{C}$.

4.4. News distributed representation optimization

The news distributed representation optimization is the adopted auxiliary task of the multi-task learning component of PMTL-DisCo. Fig. 2 shows the subgraphs of the veracity dissemination consistency network constructed from the two benchmarks, **PolitiFact** and **GossipCop**. It confirms the theory of "user spreading news engagement bias", which is applicable to the phenomenon of news reposting veracity consistency (*i.e.*, "news veracity dissemination consistency": Given that this network contains a wealth of information about whether there are

reposting relationships between news and the frequency of reposting, this characteristic is closely related to the judgment of news veracity.)

The major distribution characteristics/features of the network is manifests in the nodes constitution and the edges relationship between nodes. Therefore, for effectively incorporating external knowledge from the dissemination network, we believe that the distribution patterns of news nodes in the network can provide helpful external bias signals for the primary task of PMTL-DisCo, prompt learning-based rumor verification task.

Thus, the auxiliary task, news distributed representation optimization, aims to enhance the rumor verification task by designing a additional learning objective function about optimizing the spatial distance (similarity) among the nodes in the veracity dissemination consistency network, which enables PLM parameters to possess valuable transfer bias from the learning of veracity dissemination consistency network for the primary task, rumor verification.

The knowledge representation learning is a ideal solution for learning network structural characteristics (Lin et al., 2018; Bengio et al., 2014), particularly the classic TransE algorithm (Translating Embedding) (Bordes et al., 2013) algorithm, a renowned distributed triple knowledge representation learning approach. During its distributed representation optimization, TransE introduces a margin ranking criterion-based objective function, as follows.

$$\mathrm{L} = \sum_{(\mathrm{h,l,t})\in \mathrm{S}} \sum_{(\mathrm{h}',\mathrm{l},\mathrm{t}')\in \mathrm{S}'_{(\mathrm{h,l,t})}} \left[\gamma + \mathrm{d}(\mathrm{h}+\mathrm{l},\mathrm{t}) - \mathrm{d}\left(\mathrm{h}'+\mathrm{l},\mathrm{t}'\right)\right]_+ . \tag{7}$$

where the $\mathrm{d}(\cdot)$ is Euclidean L2-norm distance, representing the spatial distance between the coordinates of two embeddings; $(\mathrm{h},\mathrm{l},\mathrm{t})$ and $(\mathrm{h}',\mathrm{l},\mathrm{t}')$ denote the positive and negative pairs from positive S and positive sets $\mathrm{S}'_{(\mathrm{h,l,t})}$; $\mathrm{d}(\mathrm{h}+\mathrm{l},\mathrm{t})$ denotes the calculated distance of the positive pair; $\mathrm{d}\left(\mathrm{h}'+\mathrm{l},\mathrm{t}'\right)$ represents the calculated distance of the negative pair; $[x]_+$ signifies the positive operation of x, and $\gamma > 0$ denotes the distance factor.

Building upon TransE, we crafted a novel algorithm derived from TransE, tailored for the distributed characteristics of the veracity dissemination consistency network, termed the news distributed representation optimization task.

Algorithm 3 illustrates the process of optimizing news distributed representation using the veracity dissemination consistency network's structural relationships associated with news nodes. It is worth noting that, compared to the prompt learning-based rumor verification task optimization described in Section 4.2, which requires 'BertForMaskedLM', news distributed representation optimization only needs to optimize the encoder of 'BertForMaskedLM', *i.e.*, the hidden state sequential output by the last layer of 'BertModel' insided in 'BertForMaskedLM'. In experiments, we take the hidden state at the position of the '[CLS]' token as the sentence-level semantic representation for the current news case.

As shown in Algorithm 3, we introduce an approach for random sampling of positive and negative samples. Specifically, we take any two neighborhood news, denoted as n_i and u_i, that have a direct reposting dissemination relationship as positive samples and randomly sample samples from the veracity dissemination consistency network that do not have a direct reposting relationship with n_i, denoted as $\neg u_i$, as negative samples for n_i. Then, we organize positive and negative samples into pairs, with positive pairs as n_i, $\neg u_i$ and negative pairs as n_i, u_i.

Drawing on the concept of TransE (Bordes et al., 2013) (See Eq. (7)), we design an optimization objective L based on the margin ranking criterion for the case distributed representation of each news case n_i, as shown in Eq. (8).

$$\begin{aligned} L_{dr}^{n_i} &= [\gamma + dis(n_i,u_i) - dis(n_i,\neg u_i)]_+, \\ &= [\gamma + Cosine(n_i,u_i) - Cosine(n_i,\neg u_i)]_+. \end{aligned} \tag{8}$$

Here, the γ represents the similarity margin, we measure the distance between news in the dissemination network using the cosine similarity metric. The entire process of loss calculation, objective integration, and parameters optimization is outlined in Algorithm 3.

Considering that the calculations of cosine similarity range between [−1, 1], where 1 represents complete similarity and −1 represents extreme dissimilarity. In experiments, we investigated the performance with γ set between [0.1, 0.9]. And we observed that when γ was set to 0.1, PMTL-DisCo can achieve promising results.

Algorithm 3 The training procedure of the news distributed representation optimization sub-task.

Input: Veracity Dissemination Consistency Network *i.e.*, Undirected Graph $\mathbf{G} = (\mathcal{N}, \mathcal{L})$ stored in Neo4j graph database;
Training Epochs N; the "BertModel" encoder from 'BertForMaskedLM', denoted as $\mathcal{PLM}$; θ denotes the parameters of $\mathcal{PLM}$;
Distance margin γ; Learning rate LR;

for all news node n_i in $\mathcal{N}$ **do**
  Collect the first-order neighboring news nodes set of n_i, denoted as U_{n_i};
end for
for $i = 1, 2, \cdots, N$ **do**
  for all news node n_i in $\mathcal{N}$ **do**
    Randomly sample the first-order neighboring news node u_i from U_{n_i};
    Randomly sample the news node $\neg u_i$ from $\mathcal{N}$ that not exists in U_{n_i};
    Fed $\mathcal{PLM}$ with the news content of $n_i, u_i, and \neg u_i$;
    Obtained the last hidden state representations $H_{n_i}, H_{u_i}, and\, H_{\neg u_i}$ at '[CLS]' position for $n_i, u_i, and \neg u_i$;
    Calculate the distance between n_i and u_i, $\leftarrow dis(n_i,u_i) =$ Cosine(H_{n_i}, H_{u_i});
    Calculate the distance between n_i and $\neg u_i$, $\leftarrow dis(n_i,\neg u_i) =$ Cosine$(H_{n_i}, H_{\neg u_i})$;
    Obtain the margin distance loss function $L_{dr}^{n_i} \leftarrow [\gamma +$ dis(n_i,u_i) - dis$(n_i,\neg u_i)]_+$;
  end for
  the aggregated loss $L_{dr} \leftarrow \sum_{n_j\in\mathcal{N}} L_{dr}^{n_j}$;
  Compute and Update $\mathcal{PLM}$ parameters by
  $\theta^* \leftarrow \theta - LR \cdot (\partial \mathcal{L}_{dr}(\theta)/\partial\theta)$
  $= \theta - LR \cdot \sum_{n_j\in\mathcal{N}} [\gamma +$ dis(n_i,u_i) - dis$(n_i,\neg u_i)]_+$;
end for

4.5. Joint optimization objective

As depicted in Fig. 3, following the multi-task learning paradigm, PMTL-DisCo simultaneously minimizes the loss functions corresponding to both the primary task and the auxiliary task during the training phase.

Firstly, for the primary task of PMTL-DisCo, "rumor verification", as described in Section 4.3.2, its training optimization loss function L_{rv} composed of standard cross-entropy loss serves as the component of the joint optimization objective during model training. Secondly, for the auxiliary task of PMTL-DisCo, "news distributed representation optimization", as described in Section 4.4, its loss function L_{dr} also serves as the component of the joint optimization objective.

As shown in Eq. (9), both of these sub-lossen L_{rv} and L_{dr} are aggregated into a unified optimization objective, L_{total}, wherein the weights are adjusted by user-defined hyperparameters α_1 and α_2.

$$L_{total} = \alpha_1 \cdot L_{rv} + \alpha_2 \cdot L_{dr}. \tag{9}$$

Due to the relatively higher abundance of training examples derived from negative sampling in the news distributed representation optimization task compared to the few-shot rumor verification task, and considering the impact of the auxiliary task on the primary task, we need to cautiously allocate weights to avoid overfitting on the

auxiliary task resulting in a decline in performance on the primary task. Therefore, we assign less weight to the α_2 of the news distributed representation optimization task. In the experiments, the default values are set to $\alpha_1 = 1$ and $\alpha_2 = 0.01$.

4.6. Multiple neighboring news reasoning enhancement strategy

During the PMTL-DisCo inference, we introduce a technique based on neighboring news reasoning (Lenz and Bergmann, 2023), called the Multiple Neighboring News Reasoning Enhancement Strategy. This strategy retrieves neighboring news through the veracity dissemination consistency network, and integrates the veracity features of these neighboring news to further optimize the detection results.

Given a news case to be predicted (denoted as $\mathbf{n}^0$), and the Veracity Dissemination Consistency Network $\mathbf{G} = (\mathcal{N}, \mathcal{L})$ stored in graph databases like Neo4j, we retrieve first-order nodes and links/relationships ($\mathbb{N}^1 = \{\mathbf{n}_0^1, \ldots, \mathbf{n}_i^1, \ldots\}; \mathbb{R}^1$, which are news), as well as second-order nodes and links/relationships ($\mathbb{N}^2 = \{\mathbf{n}_0^2, \ldots, \mathbf{n}_i^2, \ldots\}; \mathbb{R}^{21}$; $\mathbb{R}^{22}$, which are also news) adjacent to the current news case, using the following database query statement: 1. MATCH (n:NC{$\mathbf{n}^0$})-[rel1:repost] → (m:NC) RETURN rel1,m;

2. MATCH (n:NC{$\mathbf{n}^0$})-[rel1:repost] →[rel2:repost] → (m:NC) RETURN rel1,rel2,m; where the "NC" denotes the type of News Case; the "repost" represents the link type; "rel1,m" in code # 1 denote the $\mathbb{N}^1, \mathbb{R}^1$; "rel1,rel2,m" in code # 2 denote the $\mathbb{R}^{21}, \mathbb{R}^{22}, \mathbb{N}^2$.

Since a node may simultaneously be a first-order and second-order node of the current news case $\mathbf{n}^0$, to prevent duplicate calculations, we deduplicate the nodes in sets $\mathbb{N}^1$ and $\mathbb{N}^2$. Specifically, we retain all nodes in the first-order node set $\mathbb{N}^1$ and remove nodes from $\mathbb{N}^2$ that appear in $\mathbb{N}^1$.

Subsequently, we calculate the trustworthiness of each news case in the de-duplicated node sets of $\mathbb{N}^1$ and $\mathbb{N}^2$. Trustworthiness reflects the confidence level in the veracity prediction results for each news case, ranging from 0 to 1, where higher values indicate greater reliability.

Given any news case from sets $\mathbb{N}^1$ and $\mathbb{N}^2$, denoted as m_i, and a trained prediction module from PMTL-DisCo for Prompt Learning-based Rumor Verification, represented as the function $P(\cdot)$, where the input is the news case and the output is its real/fake label, *i.e.*, (0 or 1); $P(\cdot)$ determines the news case veracity by simply selecting the category corresponding to the highest probability value. $B_2 I(\cdot)$ is a function that converts boolean values to integers, *i.e.*, Bool type to Integer type. The calculation process of the trustworthiness β_{m_i} of each news case is as follows.

Similarly, we first obtain the first-order and second-order neighboring news case sets of news case m_i using Code # 1, denoted as $\mathbb{M}^1$ and $\mathbb{M}^2$, respectively. Following that, we derive the trustworthiness of news case m_i through the following formula:

$$\beta_{m_i} = \frac{\sum_{m_j \in \mathbb{M}^1 \cup \mathbb{M}^2} B_2 I(P(m_i) == P(m_j))}{|\mathbb{M}^1| + |\mathbb{M}^2| - 1}. \tag{10}$$

where the $|\mathbb{M}^1|$ and $|\mathbb{M}^2|$ respectively denote the length of $\mathbb{M}^1$ and $\mathbb{M}^2$.

After obtaining the trustworthiness β_{m_i} of each news case m_i in $\mathbb{N}^1$ and $\mathbb{N}^2$, we calculate the influence weight δ_{m_i} of each neighboring news case m_i in $\mathbb{N}^1$ and $\mathbb{N}^2$ on the current news case to be detected through the following steps:

$$\gamma_{m_i} = \begin{cases} \mathbf{R^1}(m_i) & if \quad m_i \in \mathbb{N}^1; \\ Min(\mathbf{R^{21}}(m_i), \mathbf{R^{22}}(m_i)) & if \quad m_i \in \mathbb{N}^2. \end{cases} \tag{11}$$

where the γ_{m_i} denotes the adopted news reposting frequency in the links/ relationships $\mathbb{R}^1, \mathbb{R}^{21}$, and $\mathbb{R}^{22}$ between the current news case $\mathbf{n}^0$ and m_i of $\mathbb{N}^1$ and $\mathbb{N}^2$; $\mathbf{R^1}(\cdot), \mathbf{R^{21}}(\cdot), \mathbf{R^{22}}(\cdot)$ denote the search function of news reposting frequency on each $\mathbb{R}^1, \mathbb{R}^{21}$, and $\mathbb{R}^{22}$, respectively.

Eq. (11) reflects the reposting frequency of neighboring news selection strategy. For second-order nodes, we consider taking the minimum reposting frequency from both the first- and second-order links, which is theoretically explainable from the decreasing entropy of news dissemination information (Lenz and Bergmann, 2023).

$$\delta_{m_i} = \frac{log(\gamma_{m_i} \times \beta_{m_i})}{\sum_{m_k \in \mathbb{N}^1 \cup \mathbb{N}^2} log(\gamma_{m_k} \times \beta_{m_k})}. \tag{12}$$

where the logarithm function $log(\cdot)$ is a mathematical operation that calculates the exponent to which a given base, typically denoted as e.

Eq. (12) illustrates that we consider both the adopted news reposting frequency γ_{m_i} and the trustworthiness β_{m_i} of news case m_i as proportional conditioning factors for the influence weight δ_{m_i} on the current news case. Through the Softmax function, the values of these products are transformed into a probability distribution, thereby forming the magnitude of influence weight.

As depicted in Fig. 3, given the current news case $\mathbf{n}^0$, through the expanded multi-label words mapping strategy in Section 4.3.2, we obtain the predicted probabilities for the real and fake categories respectively as $p_+(\mathbf{n}^0)$ and $p_-(\mathbf{n}^0)$.

Then, we set a threshold K, and for news where $|p_+(\mathbf{n}^0) - p_-(\mathbf{n}^0)| > K$, we default to selecting the category with the higher predicted probability as the final FEND prediction. In our experiments, setting $K = 0.35$ yields optimal performance. For news where $|p_+(\mathbf{n}^0) - p_-(\mathbf{n}^0)| < K$, we incorporate a multiple neighboring news reasoning enhancement, which leverages news with higher dissemination consistency to further refine predictions.

The specific process for refining the predictions of the news case $\mathbf{n}^0$ is as follows:

$$\begin{aligned} &\left[\hat{p}_+(\mathbf{n}^0); \hat{p}_-(\mathbf{n}^0)\right] = \mu \times \left[p_+(\mathbf{n}^0); p_-(\mathbf{n}^0)\right] \\ &+(1-\mu) \sum_{m_i \in \mathbb{M}^1 \cup \mathbb{M}^2} \left(\left[p_+(\mathbf{m}_i); p_-(\mathbf{m}_i)\right] \times \delta_{m_i} \right). \end{aligned} \tag{13}$$

where the $[\hat{p}_+(\mathbf{n}^0); \hat{p}_-(\mathbf{n}^0)]$ are the refined predictions of news case $\mathbf{n}^0$. We additionally set a hyperparameter μ ranging from 0 to 1, reflecting the extent to which neighboring news influence the original predicted values. In our experiments, we chose $\mu = 0.5$, and thus achieved relatively good results.

Leveraging the neighboring news from the veracity dissemination consistency network based on Eq. (13), we further adjust the labels probability distribution of uncertain news case. We then select the category corresponding to the adjusted probability with the highest value as the final real/fake prediction label, thereby enhancing the performance of the FEND task.

5. Experiments

5.1. Experimental setup

5.1.1. Datasets and statistics

We evaluate our PMTL-DisCo framework using three widely-adopted real-world benchmark datasets: FakeNewsNet (Shu et al., 2019b), which comprises the PolitiFact and GossipCop datasets, and the FANG (Nguyen et al., 2020) dataset. These datasets are commonly utilized in existing work for fake news detection research. They include news articles collected from reputable fact-checking websites, alongside corresponding social user engagement data retrieved from Twitter, including repost user IDs.

The FakeNewsNet dataset (Shu et al., 2019b) is a comprehensive repository designed to support fake news research by incorporating diverse features, including news content, social context, and spatiotemporal information. The PolitiFact and GossipCop datasets, which form part of FakeNewsNet, have been constructed using an end-to-end system called FakeNewsTracker. PolitiFact contains fact-checked news articles categorized into real and fake news, accompanied by social engagement data such as user comments and reposting activities. Similarly, GossipCop focuses on celebrity gossip and includes news with similar annotations and social signals. FakeNewsNet provides a multi-dimensional data repository that facilitates the study of fake news detection, diffusion, and intervention.

Table 3
Dataset statistics for the adopted benchmarks, PolitiFact, GossipCop, and FANG.

Dataset	# News articles	# Real news	# Fake news	# User engagements	# Distinct users
PolitiFact	482	241	241	236,950	148,023
GossipCop	1736	868	868	163,532	79,474
FANG	664	332	332	50,549	37,165

The FANG dataset (Factual News Graph) (Nguyen et al., 2020) adopts a graph representation framework to model the social context of news dissemination. Unlike prior models, FANG emphasizes representation learning by capturing social actors and their interactions in a graphical format. This dataset includes heterogeneous relationships, such as user-news stances and source-news publications, enabling the study of social engagement patterns and the development of scalable, graph-based fake news detection approaches. FANG has demonstrated robust performance in scenarios with limited training data, making it an ideal benchmark for few-shot learning settings.

To ensure consistency and comparability, we adopt the same data partitioning method as previous studies. Specifically, we split each dataset into training and test sets by randomly sampling n news items as training data ($n \in \{16, 32, 64, 128\}$), maintaining a 1:1 ratio of fake news to real news within the training data. This approach allows us to rigorously evaluate PMTL-DisCo under various few-shot settings.

The key statistics for PolitiFact, GossipCop, and FANG datasets are summarized in Table 3. The User Engagements attribute refers to the total number of interactions (such as reposts, likes, comments, or shares) that users have with news articles, capturing the extent of social media activity related to each dataset. Meanwhile, the Distinct Users attribute represents the unique number of individuals engaging with these articles, reflecting the breadth and diversity of audience interactions. Together, these attributes provide valuable insights into the dissemination patterns and user behaviors surrounding fake and real news, aiding the analysis of social context.

5.1.2. Baselines

We assess our PMTL-DisCo against several prominent baselines, each representing "Train-from-Scratch" approaches (employing FEND task-specific neural architectures) and "PLM-based" approaches (aiming to mitigate label scarcity through rich pre-trained knowledge in PLMs).

"Train-from-Scratch" baseline methods include:

1. **dEFEND-variant** (Shu et al., 2019a), a modified version of the hierarchical attention model dEFEND;
2. **SentGCN** is graph-based methodologies leveraging Graph Convolutional Network (GCN);
3. **SentGAT** (Vaibhav et al., 2019) is graph-based methodologies leveraging Graph Attention Network (GAT);
4. **SAFE-variant**, a variant of SAFE that excludes the visual component and incorporates a TextCNN-based module to process news article texts;
5. **GCNFN** employs deep geometric learning to model news dissemination patterns alongside textual node embedding features;
6. **GraphSAGE** (Nguyen et al., 2020) constructs a heterogeneous social graph comprising news articles, sources, and social users for fake news detection.

"PLM-based" baseline methods include:

1. **BERT-FT** (Devlin et al., 2019) utilize RoBERTa alongside a task-specific MLP to predict the veracity of news articles;
2. **RoBERTa-FT** (Devlin et al., 2019) utilize BERT alongside a task-specific MLP to predict the veracity of news articles;
3. **PET** (Schick and Schütze, 2021) provides task descriptions to PLMs for supervised prompt-tuning via task-related cloze questions and verbalizers;
4. **KPT** (Hu et al., 2022a) broadens the label word space with class-related tokens of varying granularities and perspectives;
5. **KPL** (Jiang et al., 2022) (Knowledgeable Prompt Learning) employs prompt learning techniques to bridge the gap between pretraining and fine-tuning by designing sophisticated prompt templates and verbal words, further enhancing its effectiveness by integrating knowledge graph entities into the prompt representation;
6. **Prompt-and-Align (P&A)** (Wu et al., 2023) is a prompt-tuning method for few-shot FEND that harnesses the pre-trained LMs and external social context topology knowledge.

To ensure consistency in experimental conditions, we choose BERT-base from Huggingface.[7] Specifically, the "BertForMaskedLM" model of "bert-base-uncased" serves as the backbone for prompt-tuning, whether in the baseline methods or our PMTL-DisCo approach.

Hyperparameters We configured the few-shot settings to {16, 32, 64, 128}, with a total of three independent training epochs per round. The batch size used for each model update was set to 8, and the maximum token sequence length for the Pre-trained Language Model (PLM) input was set to 128. We set the default random seed to 0 and the learning rate to 0.00005. The hyperparameters threshold K and μ were set to default values of 0.35 and 0.65, respectively, which were determined as the optimal settings through experimentation. During computing the trustworthiness β_{m_i} of news case m_i, the function $P(\cdot)$ directly selects the label words with higher predicted probabilities to determine the labels. The chosen evaluation metric is accuracy, which measures the ratio of correctly predicted news articles to the total predicted cases, with a higher ratio indicating better performance.

5.2. Main results

We tested our PMTL-DisCo against other baselines for accuracy on the PolitiFact, GossipCop, and FANG datasets, as shown in Table 4. We classified the baselines into two groups according to the "Train-from-Scratch" and "PLM-based" categories of FEND

From this table, it is clear that compared to these baselines, our PMTL-DisCo surpasses all methods in accuracy, validating the effectiveness and superiority of PMTL-DisCo that we propose. This demonstrates the necessity of introducing the expanded label words-based verbalization engineering, the "news distributed representation optimization" auxiliary task, and the "multiple neighboring news reasoning enhancement" strategy into the original prompting-based FEND method. This also indirectly reflects the significance of incorporating the "news veracity dissemination consistency" phenomenon into the few-shot FEND task.

Moreover, it can be seen that for the first group using methods based on task-specific neural architecture, their prediction performance depends heavily on the size and quality of the supervised data. For example, for GCNFN, the performance on the **PolitiFact** dataset with 16-shot is only 55.37%, compared to an accuracy of 84.46% with 128-shot, resulting in a significant decrease of approximately 29.09% in accuracy.

For the FANG dataset, the veracity feature pertaining to news authenticity detection is intricately linked with the interrelations among three distinct node categories: users, news articles, and media sources. Relative to analogous datasets such as PolitiFact and GossipCop, the reliance on semantic information derived from the content of news sentences is comparatively subdued. Concurrently introduced with

[7] Huggingface: https://huggingface.co/google-bert/bert-base-uncased *[Accessed on 2024.03]*.

Table 4
Performance comparisons (% Accuracy) of our PMTL-DisCo and baselines on three benchmarks. Some baseline statistics are from their respective literature (Wu et al., 2023). Bold numbers signify the best performance, underlined indicate second-best. Results marked with * are enhanced using "multiple neighboring news reasoning.".

FEND approaches	PolitiFact				GossipCop				FANG			
	16	32	64	128	16	32	64	128	16	32	64	128
dEFEND-variant	51.78	54.62	61.06	66.34	50.43	50.35	51.48	52.66	50.19	50.85	50.91	54.60
SentGCN	56.34	51.95	56.81	56.23	49.57	49.33	50.04	53.87	51.35	50.60	52.58	54.45
SetnGAT	56.09	52.99	55.64	58.36	49.54	50.11	50.52	54.65	51.04	51.59	54.09	56.34
SAFE-variant	57.71	54.52	61.91	68.36	51.35	51.20	53.39	57.40	51.90	53.01	55.53	58.35
GraphSAGE	57.87	60.45	68.72	77.56	51.73	54.10	63.93	69.65	53.32	56.22	58.53	60.44
GCNFN	55.37	61.86	70.06	84.46	52.18	53.85	62.29	71.89	52.43	56.08	60.12	61.85
BERT-FT	61.27	67.48	73.52	77.43	52.45	52.74	55.02	59.28	53.28	54.71	57.02	58.41
RoBERTa-FT	54.80	61.16	78.95	81.44	52.54	54.17	54.09	61.35	51.47	54.56	56.92	60.92
PET	64.24	68.09	79.44	80.49	53.69	55.11	59.78	62.97	55.76	56.46	59.13	59.79
KPT	68.27	70.21	80.40	83.16	53.95	54.70	60.12	62.19	56.87	55.79	60.36	61.38
KPL (Knowledge-able Prompt Learning)	74.60	79.73	81.44	84.26	41.63	46.92	48.83	52.74	57.88	58.47	62.09	65.81
Prompt-and-Align (P&A)	85.30	83.73	86.79	89.45	61.15	70.59	74.62	81.60	59.42	60.48	62.37	64.37
PMTL-DisCo	**89.04**	**90.12**	**91.84**	**92.99**	**69.29**	**76.42**	**83.55**	**84.45**	**65.81**	**69.72**	**72.29**	**73.14**

FANG, the GraphSAGE methodology attained an accuracy of 75.18% under full-supervised learning conditions, where data volume is ample. Consequently, conventional "Train-from-Scratch" and "PLM-based" approaches failed to manifest the anticipated predictive prowess within small-sample scenarios. Evidently, juxtaposed against these benchmarks, our proposed approach leverages a social context veracity dissemination consistency network, which integrates a suite of dissemination consistency-guided reasoning strategies. This integration, coupled with the optimization of node features through multi-task learning and prediction stages, achieved performance approaching that of fully supervised methods on the FANG dataset, *e.g.*, 69.72% in 32-shot and 73.14% in 128 shot. This advancement heralds a promising development with profound practical implications.

The comparison between PMTL-DisCo and KPL highlights the superior performance of our proposed model across all evaluated metrics and datasets. PMTL-DisCo consistently outperforms KPL, particularly in challenging scenarios with limited resources, demonstrating its robustness and effectiveness. For example, PMTL-DisCo achieves with scores ranging from 89.04 to 92.99, compared to KPL's range of 74.60 to 84.26. Similarly, on more resource-rich datasets, GossipCop, PMTL-DisCo maintains its lead with accuracy scores 69.29 to 84.45, compared to KPL's 41.63 to 52.74. This further demonstrates PMTL-DisCo's superior robustness and performance, leveraging its advanced multi-task learning and reasoning strategies.

Additionally, the statistical results indicate that the "PLM-based" method also significantly outperforms the "Train-from-Scratch" method, confirming the effectiveness of leveraging the rich pre-trained knowledge in PLMs to mitigate label scarcity. Such methods are more suitable for few-shot FEND tasks than devising task-specific neural architectures for fake news detection. Furthermore, the comparison between Prompt-And-Align, KPT, and PET methods against simple fine-tuning methods on BERT/RoBERTa intuitively demonstrates that the prompt learning paradigm can better stimulate the prior linguistic knowledge learned during pre-training in PLMs, thereby achieving better performance than the fine-tuning paradigm.

In conclusion, through experiments, it firmly proves that the "news veracity dissemination consistency" phenomenon can provide an important indication of news veracity consistency for the fake news detection task. The veracity dissemination consistency reasoning can serve as crucial indicative external knowledge for the task of fake news detection, thereby guiding the original FEND model to achieve better predictions.

5.3. Ablated studies

In this section, we discuss the effectiveness of each module of our PMTL-DisCo and the rationale behind the selection of hyperparameters through a series of comparative experiments.

5.3.1. Effect of dissemination consistency-guided reasoning augmentation

As delineated in the preceding discussion, our PMTL-DisCo seamlessly integrates dissemination consistency-guided reasoning augmentation within both the training/optimization and inference/prediction phases. Specifically, it introduces a multi-task learning paradigm onto the prompt-tuning-based rumor verification classifier, concomitantly embracing the auxiliary task of adopted news distributed representation optimization alongside the primary task. Furthermore, it incorporates a sophisticated strategy for enhancing reasoning using multiple neighboring news articles. Leveraging the veracity dissemination consistency network, this strategy adeptly retrieves neighboring articles and synthesizes their contextual information to meticulously rectify the initial FEND prediction outcomes.

Correspondingly, as delineated in Table 5, we systematically excluded the multi-task learning paradigm in training/optimization phase, the prediction refinement strategy in inference/prediction phase, and both simultaneously from the initial PMTL-DisCo. This yielded three variants of PMTL-DisCo, which are compared against the original PMTL-DisCo (our default complete method in the fourth column), denoted respectively as: **Prompting** in the first column; + **MTL** in the second column; and + **Reason Enhance** in the third column. Likewise, comprehensive comparative experiments were conducted across three benchmarks and four different few-shot scenarios for all variants. From this table, results underscored with underscores represent the baseline, i.e., **Prompting**: the original prompting-based PLM solution without any dissemination consistency-guided reasoning augmentation. Performance improvements, highlighted in green, indicate the enhanced effectiveness of our PMTL-DisCo variants with reasoning augmentation compared to **Prompting**.

From Table 5, it can be observed that the aforementioned three variants significantly improve the performance of the original PLM solution based on **Prompting**. Additionally, evaluations across three benchmarks reveal that + MTL can boost overall prediction accuracy by 3% to 10% absolute performance gain, while the improvement brought by + **Reason Enhance** compared to + **MTL** is even more significant. Although both strategies fundamentally integrate external knowledge of veracity dissemination consistency network's new reposting relationship into the initial prompting learning framework of PMTL-DisCo, these approaches achieve this through different ways: one provides supervised signals oriented towards rumor text for the model itself, while the other achieves news veracity feature integration by merging relationships between news case nodes in the veracity dissemination consistency network. It can be inferred that veracity consistency among the news nodes of dissemination network has a more pronounced effect on rumor detection improvement.

It is worth noting that in specific cases, such as FANG with 128-shot setting, the combined use of + **MTL** and + **Reason Enhance** does not outperform using either module individually. After revising

Table 5
The ablation comparisons concerning the impacts of dissemination consistency-guided reasoning augmentation in our PMTL-DisCo.

PMTL -DisCo Variants		Prompting (*Baseline*)	+ MTL	+ Reason Enhance	PMTL -DisCo (Default)
Politi -Fact	**16**	79.38	86.79 ↑ 7.41%	88.70 ↑ 9.32%	89.04 ↑ 9.66%
	32	79.67	86.82 ↑ 7.15%	88.92 ↑ 9.25%	90.12 ↑ 10.45%
	64	81.49	88.70 ↑ 7.21%	89.37 ↑ 7.88%	91.84 ↑ 10.35%
	128	82.48	89.45 ↑ 6.97%	91.84 ↑ 9.36%	92.99 ↑ 10.51%
Gossip -Cop	**16**	60.67	61.63 ↑ 0.96%	65.28 ↑ 5.15%	69.29 ↑ 8.62%
	32	66.00	70.13 ↑ 4.13%	73.74 ↑ 7.74%	76.42 ↑ 10.42%
	64	69.29	79.66 ↑ 10.37%	80.91 ↑ 11.62%	83.55 ↑ 14.26%
	128	72.37	81.00 ↑ 8.63%	83.84 ↑ 11.47%	84.45 ↑ 12.08%
FANG	**16**	58.77	63.02 ↑ 4.25%	61.37 ↑ 2.60%	65.81 ↑ 7.04%
	32	60.23	65.93 ↑ 5.70%	64.12 ↑ 3.89%	69.72 ↑ 9.49%
	64	63.22	70.48 ↑ 7.26%	68.70 ↑ 5.48%	72.29 ↑ 9.07%
	128	64.98	75.29 ↑ 10.31%	74.31 ↑ 9.33%	73.14 ↑ 8.16%

the experimental configuration, discussing the results with other researchers, and carefully analyzing the experimental logs, we confirm that there were no errors in the experimental process. This phenomenon can be attributed to the inherent variability in few-shot experimental setups, where the limited number of samples increases susceptibility to random fluctuations. These are considered to fall within an acceptable range and do not indicate any fundamental issue with PMTL-DisCo's architecture.

Furthermore, from the fourth column of the table, it can be observed that PMTL-DisCo combines the aforementioned two different approaches without affecting their respective performances. PMTL-DisCo amalgamates these distinct approaches seamlessly, leveraging their complementary discriminative capabilities to further bolster overall model performance.

5.3.2. Experimental validation with PolitiFact-Snopes-2024 dataset

To further validate the effectiveness of our proposed PMTL-DisCo framework, we introduce the PolitiFact-Snopes-2024 (Niu et al., 2024) dataset. This dataset was manually curated from PolitiFact[8] and Snopes,[9] comprising approximately 1200 verifiable claims with fact-check rating labels. Non-text-based claims and exclusive fact-checks with organization-specific supporting sources were filtered out. Due to the dataset's rich social context, it complements the external veracity dissemination network framework. Additionally, as this dataset aligns with the VeraCT Scan paradigm, we leveraged social context information originally from PolitiFact to enhance this dataset's external veracity dissemination network.

Given the origin of PolitiFact-Snopes-2024 from the VeraCT Scan paradigm, we adopt the veracity dissemination network framework derived from PolitiFact's social context information. This integration allows us to simulate a broader veracity dissemination network and enhance cross-case feature learning.

The experimental results, presented in Table 7, highlight distinct performance trends across the three models. The baseline, VeraCT Scan, shows a steady improvement as the number of few-shot samples increases, demonstrating its capability to leverage larger data sizes. However, it struggles to handle limited samples effectively, achieving an accuracy of 70.85% in the 16-shot setting and peaking at 82.15% in the 64-shot scenario. These results suggest that while VeraCT Scan can capitalize on increased data availability, its performance in low-resource settings is relatively limited. In contrast, Prompt-and-Align (P&A) consistently outperforms VeraCT Scan across all few-shot scenarios, starting at 81.12% accuracy in the 16-shot setting and improving to 84.73% at 64 shots. This trend reflects the strength of its alignment strategy and its effective use of external knowledge, which significantly boosts performance compared to the baseline.

PMTL-DisCo, however, achieves the best performance among the three models, consistently delivering the highest accuracy in all settings. Beginning at 82.03% in the 16-shot scenario, it demonstrates superior generalization and robustness, culminating in 86.42% accuracy with 64 shots. This exceptional performance highlights the benefits of PMTL-DisCo's multi-task learning approach, which incorporates veracity dissemination reasoning augmentation strategies to enhance few-shot fake news detection.

Overall, PMTL-DisCo shows a clear advantage in low-resource settings, significantly outperforming both VeraCT Scan and Prompt-and-Align (P&A) across all few-shot configurations. This experiment demonstrates the superiority of PMTL-DisCo in addressing the challenges of few-shot fake news detection. The significant improvements over both VeraCT Scan and P&A across varying few-shot settings affirm the robustness of PMTL-DisCo.

5.3.3. Influence of news embedding optimization strategy

As discussed in Section 4.4, we introduced the semantic distance measurement based on TransE (Bordes et al., 2013) and cosine similarity to realize the auxiliary task of "news distributed representation optimization". However, the design of different auxiliary tasks may introduce fluctuations in the learning effectiveness of PMTL-DisCo.

[8] PolitiFact: https://www.politifact.com/ *[Accessed on 2024.10]*.
[9] Snopes: https://www.snopes.com/ *[Accessed on 2024.10]*.

Table 6
The performance comparisons of PMTL-DisCo under various implementations of the auxiliary task, news distributed representation optimization.

PMTL-DisCo Variants		PolitiFact			GossipCop			FANG		
		16	32	64	16	32	64	16	32	64
Comparative Embedding Optimization Strategies	L1-Distance Margin	86.79 ↓ 2.25%	88.70 ↓ 1.42%	89.51 ↓ 2.33%	65.28 ↓ 4.01%	73.92 ↓ 2.51%	80.36 ↓ 3.19%	62.39 ↓ 3.42%	64.27 ↓ 5.45%	69.72 ↓ 3.20%
	L2-Distance Margin	87.52 ↓ 1.52%	88.92 ↓ 1.20%	89.04 ↓ 2.80%	66.67 ↓ 2.62%	71.38 ↓ 5.04%	81.81 ↓ 1.74%	64.29 ↓ 1.52%	67.90 ↓ 1.82%	69.72 ↓ 3.20%
	Binary Classifying	88.48 ↓ 0.56%	90.56 ↑ 0.44%	92.06 ↑ 0.22%	68.93 ↓ 0.36%	73.79 ↓ 2.63%	83.84 ↑ 0.29%	66.89 ↑ 1.08%	68.13 ↓ 1.59%	73.31 ↑ 0.39%
PMTL-DisCo (Default)	Cosine Similarity	89.04	90.12	91.84	69.29	76.42	83.55	65.81	69.72	72.92

Table 7
Few-shot performance results (Accuracy %) on PolitiFact-Snopes-2024 dataset.

Model	16-shot	32-shot	64-shot
VeraCT Scan	70.85	76.34	82.15
Prompt-and-Align (P&A)	81.12	83.25	84.73
PMTL-DisCo	**82.03**	**85.67**	**86.42**

Therefore, we cannot guarantee that the learning approach of the auxiliary task designed as in Section 4.4 will necessarily achieve optimal performance.

To further analyze the implementation details of the auxiliary task, we additionally introduced two different distance metrics as the measurement indicators for representation optimization, including the Manhattan distance based on the L1-norm and the Euclidean distance based on the L2-norm. Furthermore, we introduced a binary classification classifier, *i.e.*, a multi-layer perceptron (MLP) based binary classifier added on top of the '[CLS]' hidden state embedding, as described in Section 4.4. Specifically, by employing a sentence-level concatenate operation to integrate the original constructed pairs and using it as the input for MLP, the optimization objective is defined as follows: inputs of predicting positive pairs as positive category and inputs of negative pairs as negative category.

Table 6 shows the performance of PMTL-DisCo model variants across the **PolitiFact**, **GossipCop**, and **FANG** benchmarks, stratified by the employment of three distinct auxiliary task strategies. Green and red markings denote absolute accuracy improvements and decreases, respectively, compared to the baseline PMTL-DisCo with default configurations, which are underscored for emphasis.

From this table, it can be observed that the news embedding optimization based on Manhattan and Euclidean distance metrics yields suboptimal outcomes, whereas optimization based on cosine similarity metric and binary classification exhibit superior performance. Consequently, in the context of selecting optimization paradigms for multi-task learning, preference is accorded to the refined TransE algorithm, leveraging cosine similarity as the metric function, alongside simplistic binary classification approaches.

Question: *Why were L1 distance, L2 distance, and cosine similarity chosen for the auxiliary task, and what are the specific roles of these metrics in news embedding optimization?*

Regarding the rationale for employing different distance metrics (L1 distance, L2 distance, and cosine similarity), different distance metrics were used in our auxiliary task design to evaluate their suitability for capturing the spatial relationships in the news embedding space, which is influenced by the properties of the veracity dissemination consistency network.

1. L1 Distance: L1 distance (Manhattan distance) is particularly effective in handling sparse and high-dimensional vectors, which are common in textual embeddings. This property is advantageous when embeddings contain many zero values, as it minimizes the effect of small, noisy deviations. This is especially useful for capturing subtle variations in sparse embeddings.

2. L2 Distance: Measures the Euclidean distance and emphasizes larger deviations due to its quadratic nature. It captures the overall magnitude or intensity of changes in embeddings, which is beneficial when the absolute scale of differences between news embeddings is important for distinguishing veracity patterns.

3. Cosine Similarity: Focuses on the angular relationship between vectors and is scale-invariant, which is crucial in scenarios where the directional alignment of embeddings is more informative than their magnitude. This property aligns with scenarios where the directional alignment of embeddings (e.g., semantic similarity) reflects user dissemination behavior better than their scale. It is particularly effective in maintaining consistency with social dissemination patterns.

By experimenting with these metrics, we aimed to evaluate their effectiveness in enhancing the auxiliary task's ability to refine veracity dissemination consistency. Among these, cosine similarity generally performed best due to its robustness in capturing scale-invariant relationships, which aligns with our theoretical understanding of user sharing patterns in the network.

5.3.4. Effectiveness of improved label words acquisition

Here, we investigate and analyze the superiority of our proposed improved approach for acquiring high-quality real/fake semantic-related label words by varying the acquisition methods.

The experimental comparisons in Table 9 are categorized into three groups based on the different label word collection strategies employed:

- 1. **Single** Label Words: Each category is mapped using only a single label word (e.g., "real" and "fake"), which is the mainstream label word mapping technique commonly utilized in prompt-tuning;
- 2. Human-Preferred Selected Label Words: Direct utilization of vocabulary words selected manually as suitable; These words align more closely with the semantic of the source labels from a human perspective. We consulted with a linguistics expert and recommended several label words that better capture the semantics of the source labels, as shown in Table 8.
- 3. Our Improved Approach for High-Quality Real/Fake Semantic-Related Label Words Acquisition, as described in Section 4.3.1.

From Table 9, the PMTL-DisCo based on selected single label words, selected human-preferred label words, and our improved label words acquisition approach correspond to *Single Word*, *Human-Preferred Selection*, and *Expanded Words Acquisition (Default)*, respectively. Furthermore, we conducted a comparative analysis of the effects of verbalizer engineering under different label words on the initial prompting learning and the subsequent integration of various veracity dissemination

Table 8
The selected expanded label words estimated by a linguistics expert.

Human-preferred selected label words	
Fake	Counterfeit, Faux, Bogus, Phony, Artificial, Sham, Imitation, Pretend, Spurious, Deceptive
Real	Authentic, Genuine, Legitimate, Actual, True, Valid, Veritable, Unaltered, Original, Substantive

Table 9
Comparisons of the variants of PMTL-DisCo that employ different strategies for acquiring real/fake semantic-related label words. Bold numbers signify the best performance, underlined indicate second-best.

PMTL-DisCo variants		PolitiFact			GossipCop			ANG		
		16	32	64	16	32	64	16	32	64
Single word	Prompting	75.43	76.91	79.23	53.92	56.83	60.35	56.21	57.01	57.19
	+ MTL	83.48	83.00	84.55	59.25	66.00	75.29	**60.36**	63.02	**69.98**
	+ Reason enhance	83.27	86.77	85.65	59.73	70.95	74.49	57.01	60.42	62.28
	PMTL-DisCo	**87.73**	**87.18**	**88.52**	**67.77**	**73.04**	**79.66**	60.05	**63.34**	69.81
Human-preferred selection	Prompting	78.34	79.23	81.90	57.45	64.56	67.36	57.99	58.25	60.36
	+ MTL	85.30	85.84	86.79	60.94	69.96	77.60	63.02	63.85	68.63
	+ Reason enhance	86.35	**88.92**	89.67	63.16	71.49	78.83	58.77	63.34	66.56
	PMTL-DisCo	**88.70**	86.47	**90.12**	**67.77**	**75.29**	**82.84**	**64.08**	**65.81**	**71.39**
Expanded words acquisition (Default)	Prompting	79.38	79.67	81.49	60.67	66.00	69.29	58.77	60.23	63.22
	+ MTL	86.79	86.82	88.70	61.63	70.13	79.66	63.02	65.93	70.48
	+ Reason enhance	88.70	88.92	89.37	65.28	73.74	80.91	61.37	64.12	68.70
	PMTL-DisCo	**89.04**	**90.12**	**91.84**	**69.29**	**76.42**	**83.55**	**65.81**	**69.72**	**72.29**

Table 10
Comparisons of PMTL-DisCo variants that employ different variants of the expanded multi-label words mapping strategy.

PMTL-DisCo variants	Prompting				Prompting + MTL				Prompting + eeason enhance				PMTL-DisCo i.e., Prompting+ MTL + reason enhance			
	Med	Max	Avg	Adapt	Med	Max	Avg	Adapt	Med	Max	Avg	Adapt	Median	Max	Avg	Adapt
PolitiFact (16)	75.57	76.30	78.44	**79.38**	80.49	85.30	85.30	**86.79**	80.04	86.36	88.56	**88.70**	83.16	86.79	**89.45**	89.04
GossipCop (16)	54.28	57.94	59.83	**60.67**	53.39	58.41	57.40	**61.63**	60.06	62.49	64.77	**65.28**	65.46	67.77	68.93	**69.29**
FANG (16)	55.53	57.82	57.50	58.77	60.12	60.44	61.38	**63.02**	57.83	59.90	59.15	**61.37**	63.94	63.37	64.99	**65.81**

consistency reasoning augmentation modules. Each row within every group in the table encompasses the following components: **Prompting**, representing the original PLM-based prompt learning solution; + **MTL**, indicating the integration of consistency reasoning augmentation during the training/optimization phase; and + **Reason Enhance**, signifying the incorporation of consistency reasoning augmentation during the inference/prediction phase.

Based on the observation from this table, it is evident that our method that utilizes the improved label words acquisition approach achieves the optimal accuracy compared to the former two approaches, single label words and human-preferred label words. Whether it is the results obtained solely through prompt-tuning prediction or those enhanced by "multiple neighboring news reasoning", our adopted label word acquisition approach is more effective in identifying label words suitable for MLM-based pseudo prompt-tuning, thereby enhancing the performance of PMTL-DisCo in fake news detection. Moreover, we discovered that the *Human-Preferred Selection* strategy notably outperforms the *Single Word* strategy, showcasing that employing multiple label words is more advantageous for model convergence and learning compared to using a single label word. Furthermore, the performance of PMTL-DisCo without multi-task learning, based on human-preferred label words, proves to be sufficiently capable of surpassing the predictive capability of PMTL-DisCo without multi-task learning based on our default label word acquisition approach, especially evident in the 32-shot and 64-shot scenarios on the **PolitiFact** benchmark. These findings strongly indicate the effectiveness of the adaptive weight allocation and multi-label words mapping strategy under multiple label words.

5.3.5. Effect of veracity dissemination consistency network size

To further investigate the relationship between the size of the *Veracity Dissemination Consistency Network* and the performance of PMTL-DisCo, we conducted experiments where varying proportions of graph links in the veracity network were randomly removed. This test was performed to understand how the model's performance changes as the network's connectivity decreases, simulating scenarios where fewer social context connections are available. Specifically, we considered four scenarios: 100% of links deleted (no social context), 75% of links deleted, 50% of links deleted, and 25% of links deleted. The experiments were conducted across three datasets (PolitiFact, GossipCop, FANG) under 16-shot, 32-shot, and 64-shot settings. The results are presented in Table 11.

The experimental results reveal a clear and consistent decline in performance as the percentage of deleted graph links increases across all datasets and few-shot settings. This pattern underscores the critical role of the *Veracity Dissemination Consistency Network* in enhancing the auxiliary task of News Distributed Representation Optimization, which is designed to complement and support the primary task of *Prompt Learning-based Rumor Verification*. The degradation in performance as the network becomes increasingly sparse highlights the importance of leveraging dissemination consistency signals.

With only 25% of the graph links removed, the model's performance experiences a slight decline but remains close to the original results achieved with full social context. For instance, on the PolitiFact dataset under the 32-shot setting, the accuracy decreases from 90.12 to 87.56, suggesting that the model retains robustness when the majority of the network connections are preserved. This result demonstrates the framework's ability to utilize partial social context effectively without significant loss in predictive performance.

However, as the proportion of deleted links increases to 75%, the performance decline becomes more pronounced. For example, on the GossipCop dataset under the 64-shot setting, accuracy drops from 83.55 to 73.42. This substantial reduction in performance highlights the necessity of a well-connected veracity network for the auxiliary task to provide meaningful support to the primary task. The results indicate that the quality and size of the network directly impact the ability of PMTL-DisCo to achieve optimal results, particularly in datasets where social dissemination patterns are essential.

Table 11
Impact of *Veracity Dissemination Consistency Network* size on PMTL-DisCo performance across three datasets under different few-shot settings.

Dataset		PMTL-DisCo (full social context)	Graph link randomly deleted (% percentage)			
			−25%	−50%	−75%	−100%
PolitiFact	16-shot	**89.04**	85.65	82.34	79.66	79.38
	32-shot	**90.12**	87.56	83.48	81.90	79.67
	64-shot	**91.84**	88.52	86.12	83.29	81.49
GossipCop	16-shot	**69.29**	66.00	63.34	61.43	60.67
	32-shot	**76.42**	71.49	70.12	67.94	66.00
	64-shot	**83.55**	79.81	75.29	73.42	69.29
FANG	16-shot	**65.81**	60.36	59.83	58.02	58.77
	32-shot	**69.72**	66.32	62.84	61.74	60.05
	64-shot	**72.29**	65.81	64.71	63.92	64.98

When all graph links are removed (100% deletion), the model no longer benefits from the auxiliary task, resulting in performance levels similar to a purely prompt-based approach without social context. For instance, on the FANG dataset under the 16-shot setting, accuracy decreases from 65.81 to 58.77. This drastic reduction emphasizes the importance of leveraging dissemination consistency signals for improving performance in few-shot scenarios. Without the veracity network, the model relies solely on the primary task, losing the external knowledge provided by social context.

Overall, this confirms the significant role of the veracity network in our proposed PMTL-DisCo. The network substantially enhances the model's ability to capture latent dissemination patterns and veracity-related features, particularly in few-shot settings where external social context provides critical supplementary knowledge. Although PMTL-DisCo retains some robustness in the absence of social context, these results demonstrate that the veracity network is indispensable for achieving the full potential of the framework.

5.3.6. Superiority of expanded multi-label words mapping

Here, we discuss the effectiveness of our proposed expanded multi-label words mapping strategy, which is based on a dynamic attention mechanism and adaptive weight allocation, hereinafter referred to as **Adapt** *i.e.,* Adaptive Allocation (default). This strategy allocates reasonable weights to different label words under each category to improve model prediction. We introduce three simple mapping strategies from label word probability distribution to categories for comparison:

- 1. **Med** *i.e.,* **Median**: Selects the category corresponding to the label word with the highest median probability value among all categories as final prediction;
- 2. **Max** *i.e.,* **Maximum**: Selects the category corresponding to the label word with the highest probability value across all categories as final prediction;
- 3. **Avg** *i.e.,* **Averaged**: Selects the category corresponding to the label word with the highest average probability value among all categories as final prediction.

As shown in Table 10, our dynamic attention mechanism and adaptive allocation-based label words mapping strategy achieves the best performance across all three benchmarks at 16-shot setting. Following it, the Averaged words mapping strategy shows the next best performance, highlighting the effectiveness of the dynamic attention mechanism and adaptive weight allocation algorithm designed in our PMTL-DisCo. We believe that the reason Averaged performs well is that its calculation incorporates features from all label words, making predictions more robust. In contrast, Maximum and Median have lower feature coverage and lack representativeness, making them more susceptible to the influence of noisy data, resulting in poor performance.

5.3.7. Selection of hyperparameters μ and K

Here, we explore the influence of hyperparameters, *i.e.*, the threshold K and the influence extent μ, on the performance of PMTL-DisCo to determine the optimal parameter values.

Firstly, regarding the influence extent μ, we set six different values for μ, {0.0, 0.25, 0.5, 0.75, 1.0}. Here, $\mu = 0.0$ corresponds to PMTL-DisCo relying entirely on the features of neighboring news from the veracity dissemination consistency network, while $\mu = 1.0$ corresponds to direct prediction through prompt learning without relying on multiple neighboring news reasoning enhancement. For the aforementioned ablation design, we conduct 16-shot experiments on three benchmarks, and we also perform the PMTL-DisCo variant without adopting the multi-task learning *i.e.*, the auxiliary "news distributed representation optimization" task, to further investigate the influence of different hyperparameters μ and K settings on PMTL-DisCo.

As shown in Fig. 5 (bottom-left), it can be observed that optimal performance is achieved when μ is between 0.4 and 0.6. This is also the reason why we chose $\mu = 0.5$ in our experiments. Secondly, we explore the influence of threshold K, on the performance of PMTL-DisCo to determine the optimal parameter values. We set five different values for K, {0.2, 0.35, 0.5, 0.65, 0.75}. This threshold K determines the extent to which the difference between the positive and negative probabilities of news affects the final FEND prediction. Specifically, for news where $|p_+(\mathbf{n}^0) - p_-(\mathbf{n}^0)| > K$, we default to selecting the category with the higher predicted probability as the final FEND prediction. As shown in Fig. 5 (bottom-right), we conducted 16-shot experiments on three benchmarks for K. It can be observed that optimal performance is achieved when K is around 0.35. This is the reason why we chose $K = 0.35$ in our experiments.

Similarly, as depicted in Fig. 5 (top-left and right), we observed that even without incorporating multi-task learning during the training phase, the rumor detection performance under the final hyperparameter settings of μ and K mirrors that of the initial PMTL-DisCo. This further emphasizes the rationale and optimality of the hyperparameter settings μ and K.

6. Case studies

To further elucidate why our PMTL-DisCo outperforms most competitive baselines, particularly in terms of the efficacy of multiple neighboring news reasoning enhancement (*i.e.*, dissemination consistency-guided reasoning augmentation) within PMTL-DisCo, we provide a concrete example from our PMTL-DisCo's experimental process on the **PolitiFact** benchmark. This example offers a visual demonstration of the multiple neighboring news reasoning enhancement strategy. It highlights the positive impact of utilizing the veracity features from adjacent news to enhance the original prediction provided by prompt-tuning of the current news case.

In Fig. 6, for a case (ID: politifact11115) with a gold label of fake (0), the LM-based pseudo prompt-tuning in PMTL-DisCo directly predicts the label words probabilities as follows: {real: 0.597, fake: 0.403}.

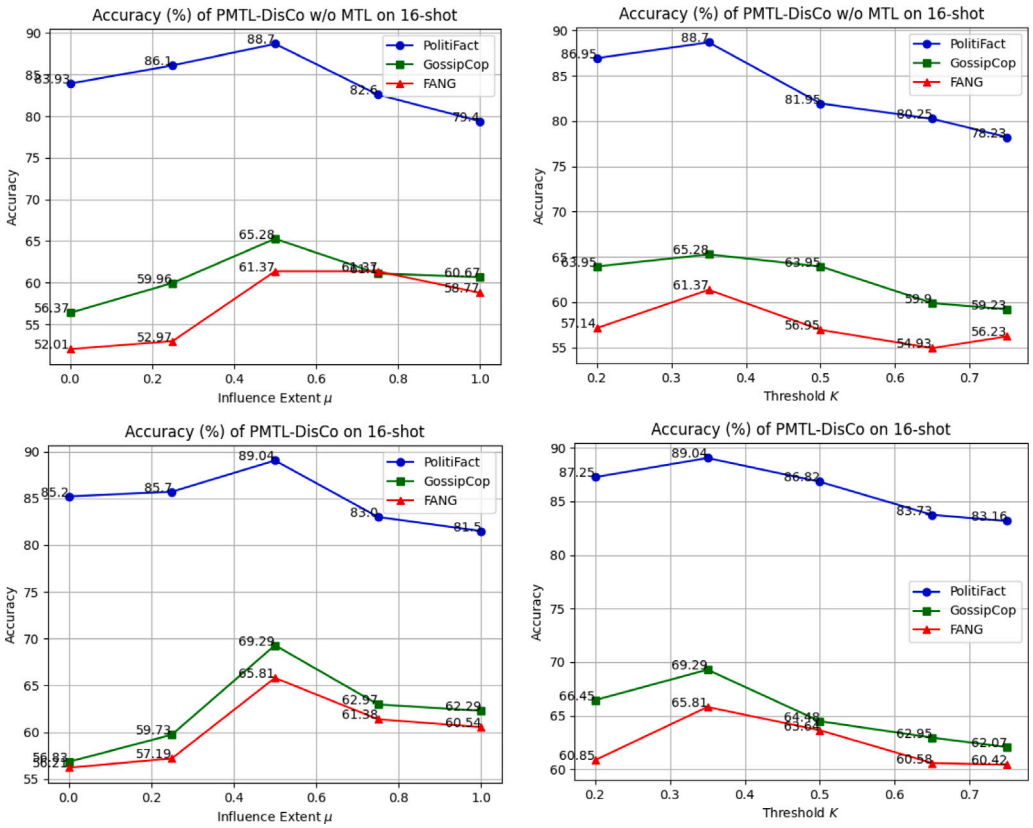


Fig. 5. The comparisons of PMTL-DisCo adopting various hyperparameter settings for μ and K (Top Sub-figures). And the comparisons of the PMTL-DisCo variant that does not adopt multi-task learning, specifically omitting the auxiliary "news distributed representation optimization" task (Bottom Sub-figures).

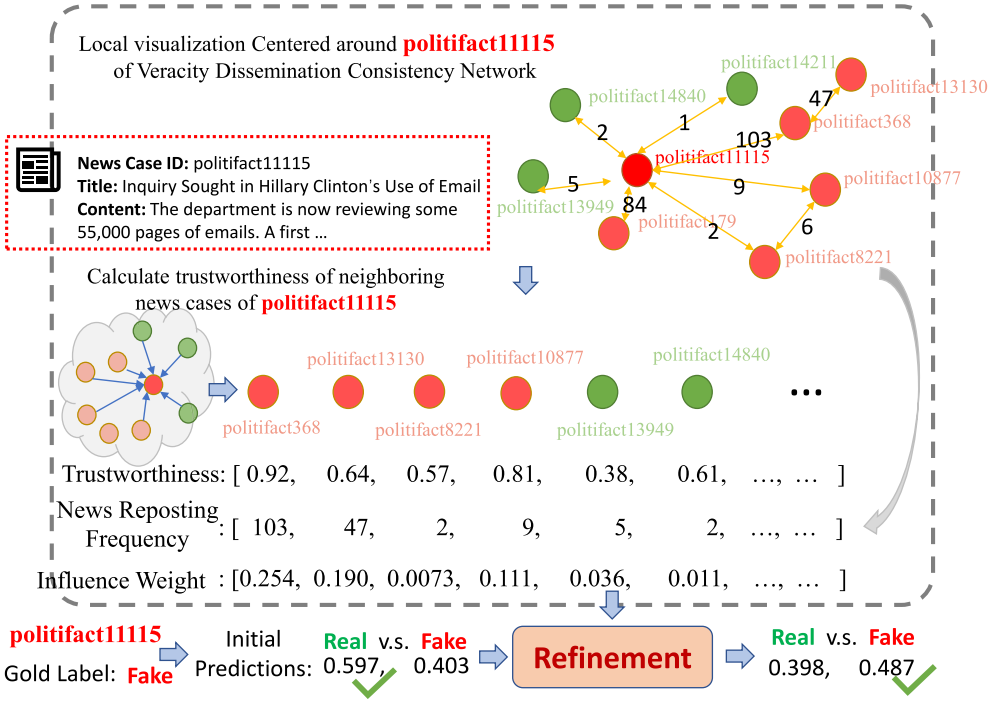


Fig. 6. The prediction optimization of social context dissemination consistency-guided reasoning of PMTL-DisCo on the news case "politifact11115" from **PolitiFact**.

Hence, the label (real) determined directly through prompt-tuning is incorrect. To refine this prediction, we employ the "multiple neighboring news reasoning enhancement" strategy. Firstly, we identify the neighboring news (first- and second-order nodes) of the "politifact11115" node in the Veracity Dissemination Consistency Network. Then, we calculate the trustworthiness and influence weight for each neighboring news case. Combining these factors with the real/fake label words probabilities predicted by the rumor verification task of PMTL-DisCo for these neighboring news, we further refine the news case "politifact11115" to obtain optimized label words probabilities: {real: 0.398, fake: 0.487}. Consequently, the predicted label of "politifact11115" is corrected to fake.

Furthermore, we conducted a statistic on similar cases to the "politifact11115" news in PolitiFact, where the initial prediction was refined accurately. The proportion of such cases is as high as 11.2% of PolitiFact test set. This significantly confirms the effectiveness of incorporating the "news veracity dissemination consistency" phenomenon into PMTL-DisCo to enhance FEND performance.

7. Discussion

7.1. Methodological comparisons with competitive baselines

In fake news detection (FEND) research, Prompt-and-Align (P&A) (Wu et al., 2023), VeraCT Scan (Niu et al., 2024), and DECOR (Wu and Hooi, 2023) stand out as highly competitive and innovative approaches, each contributing significantly to this critical problem. These methods showcase diverse methodologies—spanning graph-based refinements, retrieval-augmented systems, and prompt-based frameworks—that significantly contribute to the field.

From a methodological perspective, our PMTL-DisCo framework distinguishes itself from these baselines by addressing specific limitations in existing approaches and targeting low-resource settings with limited labeled data. Unlike other works such as DECOR (Wu and Hooi, 2023) and VeraCT Scan (Niu et al., 2024), which operate in a fully supervised setting with abundant labeled data, PMTL-DisCo achieves competitive performance using only limited samples (e.g., 16, 32, or 64 per class). This fundamental difference reflects the unique focus of PMTL-DisCo on delivering robust fake news detection solutions in low-resource environments, where extensive labeled data may not be readily available

Comparison with DECOR: DECOR employs graph neural networks (GNNs) with degree-corrected social graph refinement to suppress noise in user-news engagement graphs. While effective in fully supervised scenarios, its reliance on abundant labeled data and structural refinement limits its adaptability in few-shot contexts. In contrast, PMTL-DisCo leverages multi-task learning (MTL) with News Distributed Representation Optimization, enabling the model to learn latent dissemination features during training. By focusing on dissemination consistency within social networks, PMTL-DisCo demonstrates a distinct advantage in handling minimal supervision while maintaining strong performance.

Comparison with VeraCT Scan: VeraCT Scan, a retrieval-augmented system, excels by using GPT-4 as its backbone and aggregating internet-wide evidence to validate news claims. This approach enhances interpretability and accuracy by combining retrieval, verification, and reasoning, albeit with substantial computational overhead. PMTL-DisCo, with its BERT-based architecture, focuses on efficient performance in resource-constrained settings. Despite using a smaller model, PMTL-DisCo achieves competitive results through innovations like Expanded Label Words-based Verbalization Engineering, ensuring adaptability and effectiveness without the high resource demands of retrieval-augmented systems.

Comparison with Prompt-and-Align: Prompt-and-Align (P&A) reformulates fake news detection as a prompt-based task, leveraging pre-trained language models (PLMs) to address label scarcity. Additionally, it aligns prompting outputs with veracity signals from a news proximity graph, offering a novel way to incorporate social context. While P&A effectively integrates social alignment, its reliance on pseudo-labeling and independent prompting steps introduces potential complexity.

In contrast, PMTL-DisCo distinguishes itself by its modular and flexible handling of social signals, particularly during inference. Unlike P&A, which tightly integrates social alignment through its news proximity graph for prediction refinement, PMTL-DisCo employs the Multiple Neighboring News Reasoning Enhancement Strategy as an optional post-prediction step. This approach leverages social signals from neighboring news articles to refine veracity predictions only when such data is available, ensuring that the model remains effective even in the absence of additional social context during deployment.

By combining prompt-based learning with dissemination-guided auxiliary tasks, PMTL-DisCo provides a unified and streamlined framework that directly improves model performance while maintaining robustness and interpretability across various contexts than P&A.

In summary, while DECOR, VeraCT Scan, and Prompt-and-Align are outstanding contributions to the field of fake news detection, PMTL-DisCo demonstrates unique strengths in tackling the challenges of few-shot learning. By focusing on multi-task learning, auxiliary representation optimization, and efficient utilization of dissemination patterns, PMTL-DisCo achieves competitive performance with minimal labeled data and computational resources. This positions our framework as a highly adaptable and practical solution in scenarios where scalability and efficiency are paramount.

7.2. Minimizing dependency on social context of PMTL-DisCo

A core strength of the PMTL-DisCo framework lies in its ability to operate effectively without complete reliance on external social context, particularly during deployment. This design addresses concerns about the practicality of fake news detection models that depend heavily on the availability of social context data. In PMTL-DisCo, the first stage integrates News Distributed Representation Optimization within the Veracity Dissemination Consistency Network to enhance the primary task of Prompt Learning-based Rumor Verification, enabling the model to learn latent veracity-related features and dissemination patterns from the training data. The second stage involves the optional Multiple Neighboring News Reasoning Enhancement Strategy, which refines initial predictions by leveraging relationships between neighboring news articles in the social graph. Thus, PMTL-DisCo strategically incorporates social context during the training phase while maintaining flexibility in the inference phase. By adopting this two-stage architecture, PMTL-DisCo effectively minimizes its reliance on external social context while maintaining robust performance. Below, we outline how PMTL-DisCo achieves this:

Social Context Integration in the Training Phase: During training, PMTL-DisCo leverages the Veracity Dissemination Consistency Network as part of the auxiliary task News Distributed Representation Optimization. This auxiliary task enhances the model's ability to learn latent veracity-related features and dissemination patterns from the social network context of the training dataset. Simultaneously, the primary task of Prompt Learning-based Rumor Verification is enriched by Expanded Label Words-based Verbalization Engineering, improving the alignment of the prompt model's predictions with the true veracity labels. Crucially, the reliance on social context is confined solely to the training stage, where it serves as an input to enhance the model's generalization and robustness in identifying fake news.

Flexible Inference Without Social Context: During inference, the Multiple Neighboring News Reasoning Enhancement Strategy serves as an optional step to refine the preliminary predictions made by the Prompt Learning-based Rumor Verification task. This strategy utilizes the relationships between neighboring news articles within the social

graph to further improve prediction accuracy. However, this step is not a critical requirement and can be omitted if social context is unavailable during deployment. The core predictions of PMTL-DisCo remain robust and effective, even without this additional refinement step.

In summary, PMTL-DisCo's design avoids strict dependency on social context during deployment, effectively addressing concerns about its applicability. This advantage makes PMTL-DisCo a highly adaptable and scalable solution for fake news detection in diverse operational settings.

7.3. Adapting the model 'PMTL-DisCo' for multi-class classification

While the focus of this work is on binary classification for fake news detection, the proposed PMTL-DisCo framework is inherently flexible and can be adapted to support multi-class classification tasks with some modifications. Achieving effective multi-class classification would require adjustments in both the model and the dataset to ensure optimal performance.

From the perspective of the model, the expanded label words-based verbalizer engineering could be extended to accommodate additional classes by expanding the label word sets for each new category (e.g., "satire", "clickbait"). This process would involve identifying high-quality label words for the new categories using techniques like synonym substitution and masked language modeling, as demonstrated for the binary case. Additionally, the multi-neighbor reasoning enhancement strategy could be adapted to aggregate veracity signals not just for binary outcomes but for the extended class labels, requiring a re-weighting mechanism that reflects the increased complexity of multi-class relationships. These changes would leverage the existing architecture while introducing greater granularity.

For the dataset, transitioning to multi-class classification would necessitate the use of datasets that include detailed annotations for multiple categories of fake news, such as "partially true", "misleading", or "propaganda". In the absence of such datasets, existing binary-labeled datasets like PolitiFact and GossipCop could be re-annotated or augmented through semi-supervised learning methods or expert-curated rules. Moreover, ensuring sufficient representation for each class would be critical, particularly in few-shot scenarios, to prevent under-represented classes from impacting overall model performance.

In summary, adapting PMTL-DisCo for multi-class classification would primarily involve expanding the verbalizer engineering to handle additional labels, refining the multi-neighbor reasoning mechanism, and employing or creating datasets with appropriate multi-class annotations. These adjustments would align the model with the demands of more granular classification tasks while retaining its core strengths in leveraging dissemination consistency and social context. Future work could explore these directions to extend the applicability of PMTL-DisCo to multi-class scenarios and provide deeper insights into the nature of misinformation.

7.4. Real-world implementation of PMTL-DisCo: Opportunities and challenges

Potential Applications in Real-World Systems: Our PMTL-DisCo has significant potential for integration into real-world systems, such as social media platforms, news aggregators, and fact-checking services. Its two-stage architecture enables scalability, where the first stage can quickly filter news articles based on content, and the second stage refines predictions using dissemination consistency reasoning. This makes it suitable for real-time applications, particularly for high-risk cases. The system's few-shot learning capability allows it to adapt to new domains, including political news, celebrity gossip, and health-related misinformation, with minimal data. Additionally, its reliance on the *Veracity Dissemination Consistency Network* makes it effective at analyzing social dissemination patterns, identifying echo chambers, and curbing the spread of misinformation.

Challenges in Real-World Deployments: Despite its promise, implementing PMTL-DisCo faces several challenges. Privacy concerns emerge when leveraging social context data, as accessing user engagement patterns may conflict with regulations such as General Data Protection Regulation (Calvi, 2024) and California Consumer Privacy Act (Tran et al., 2024). Computational complexity also poses a problem, especially during the reasoning enhancement stage, which may be resource-intensive for large-scale platforms. The dynamic nature of social networks adds another layer of difficulty, as misinformation tactics and user behavior evolve rapidly, necessitating frequent model updates. Further, generating labeled training data for new domains can be costly and subjective. Interpretability is another concern—while PMTL-DisCo achieves high accuracy, its reliance on latent representations may limit transparency in decision-making. Finally, adversarial actors could attempt to manipulate social dissemination patterns to deceive the system, requiring robust safeguards against such attacks.

Balancing Complexity and Scalability in PMTL-DisCo: The complexity of the PMTL-DisCo framework arises from its multi-component design, which is intended to achieve high performance, particularly in low-resource scenarios. To balance this complexity with scalability and practical applicability, the framework is designed with a modular architecture. During the training phase, the model leverages the **Veracity Dissemination Consistency Network** as part of the auxiliary task, News Distributed Representation Optimization, to capture dissemination patterns and latent veracity-related features. In the inference phase, PMTL-DisCo focuses primarily on the Prompt Learning-based Rumor Verification task, which operates independently of auxiliary inputs. The Multiple Neighboring News Reasoning Enhancement Strategy, while available to refine predictions, is entirely optional. This modularity allows the framework to cater to diverse application requirements while maintaining its high performance and adaptability.

To summarize, implementing PMTL-DisCo in real-world fake news detection systems offers transformative potential by combining robust reasoning capabilities with the ability to adapt to diverse social contexts. overcoming challenges such as privacy concerns, computational complexity, and evolving misinformation tactics will require a multidisciplinary approach involving advancements in AI, privacy-preserving techniques, and real-time system optimization.

8. Conclusion

In few-shot FEND task, the weakness of supervised learning methods lies in the requirement of a large amount of labeled data. However, the construction of high-quality datasets is time-consuming and labor-intensive. In contrast, prompt learning technology has demonstrated superiority. Moreover, the theory of "user spreading news engagement bias" reveals the impact of social context on news dissemination, providing valuable insights. Inspired by this theory, we propose PMTL-DisCo, a prompting multi-task learning framework. It overcomes the limitations of previous supervised learning by incorporating the external knowledge of news veracity dissemination consistency with the prompt-tuning technique, achieving improved detection accuracy in data-scarce scenarios. Our PMTL-DisCo represents a significant advancement in the research of few-shot fake news detection. Moreover, the introduction of the expanded label words-based verbalizer engineering further refines the understanding capability of our PMTL-DisCo. The adoption of the novel multiple neighboring news reasoning enhancement strategy further optimizes detection performance by leveraging the neighboring news from the veracity dissemination consistency network. Through extensive experimentation, our PMTL-DisCo has been validated to effectively enhance the fake news detection capability in data-scarce scenarios. In conclusion, PMTL-DisCo offers a powerful solution for combating misinformation, ensuring the integrity of information dissemination in the digital age.

Table A.12
Comprehensive glossary of technical terms related to fake news detection and veracity dissemination consistency in PMTL-DisCo framework.

Term	Definition
Confirmation bias	Tendency of individuals to favor information aligning with their beliefs, disregarding conflicting evidence.
User spreading news engagement bias theory	Theory suggesting that users distribute news aligning with their pre-existing beliefs, affecting veracity dissemination.
News distributed representation optimization	Auxiliary task in PMTL-DisCo that optimizes the representation of news nodes within a veracity dissemination consistency network to enhance the primary task of rumor verification.
Veracity dissemination consistency network	A network graph where nodes represent news articles and edges represent shared reposting actions, used to model consistency in news veracity.
Multiple neighboring news reasoning enhancement	Strategy to incorporate veracity reasoning of neighboring news from a dissemination network for improved detection outcomes.
Expanded multi-label words mapping	A technique that dynamically assigns weights to multiple label words under a category, improving verbalization and prediction accuracy.
Dynamic attention mechanism	Allocates weights adaptively across multiple label words in label mapping strategies to enhance prediction stability.
WordNet SynSet substitution	Utilizing WordNet's interconnected synonym sets (SynSets) to expand semantic label word coverage for real/fake classification.
Word embedding substitution	Using embedding methods like Word2Vec and GloVe to retrieve semantically similar words as synonyms for enhanced label acquisition.
High-quality label words acquisition	The process of selecting precise label words from external knowledge bases (e.g., WordNet, embeddings) for enhanced verbalization in linguistic tasks.
Expanded label words-based verbalizer engineering	Approach to enhance model generalization by dynamically expanding and mapping label words in verbalization tasks, improving downstream task compatibility.
Cosine similarity optimization	Metric used in auxiliary task optimization, ensuring enhanced similarity in embedding spaces for better performance on the primary task.
MLM-based masked word substitution	Technique employing Masked Language Models (e.g., BERT) to identify and substitute high-probability words for masked positions, aiding in enhanced semantic label word acquisition.
Prompt learning	Utilizing pre-trained language model prompts for task-specific learning with minimal data.
Pseudo prompt-tuning	Fine-tuning pre-trained models using masked tokens for binary classification tasks (e.g., real/fake detection).
Multi-Task Learning (MTL)	Machine learning approach to learn multiple related tasks simultaneously to enhance model generalization.
Few-shot learning	Learning methodology where a model is trained with a limited number of labeled samples, relying on auxiliary knowledge for task performance.

CRediT authorship contribution statement

Weiqiang Jin: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ningwei Wang:** Writing – review & editing, Supervision, Investigation. **Tao Tao:** Funding acquisition. **Mengying Jiang:** Validation. **Yebei Xing:** Validation. **Biao Zhao:** Funding acquisition. **Hao Wu:** Funding acquisition. **Haibin Duan:** Funding acquisition. **Guang Yang:** Writing – review & editing, Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix. Key technical terms in PMTL-DisCo for enhanced understanding

To facilitate better comprehension and accessibility for future researchers, we have compiled a comprehensive glossary of key terms introduced in this study, as illustrated in Table A.12. This glossary serves as a quick reference guide, aiming to clarify newly introduced concepts and technical terminology that may otherwise pose challenges to understanding the content. By aligning the explanations with the context and applications within this work, we hope to make the core ideas more intuitive and actionable for readers.

Data availability

Full code to replicate the experiments are available from Github: https://github.com/albert-jin/PMTL-DisCo. The used datasets will be made available on request.

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